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Tax treaties and foreign direct investment: a network approach

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Abstract Multinational investors often reduce tax on dividends by using indirect investment routes. This paper constructs a tax rate matrix to represent a real-world network of tax treaties between 70 countries and develops a computation algorithm to study the structure of tax-minimizing (direct or indirect) investment routes in the tax treaty network. The treaty shopping rate, defined as the difference between the foreign tax rates of the direct route and a tax-minimizing route, is about 3.66 percentage points on average. This paper also examines the relationship between FDI and the structure of tax-minimizing routes. Empirical results show that the existence of a tax-minimizing direct route is positively and significantly related to FDI. The inward FDI stock via a tax-minimizing direct route is about 2.14 times larger than the inward FDI stock via a direct route that is not tax-minimizing. By making a direct route tax-minimizing, countries can encourage FDI via the direct route and reduce treaty shopping.

Keywords Tax treaty network · Tax-minimizing route · Treaty shopping rate · Foreign direct investment

JEL Classification F23 · H25 · H87

1 Introduction

The main purpose of tax treaties is to avoid, or at least to reduce, double taxation on cross-border economic activity, such as foreign direct investment (FDI). By reducing double taxation, tax treaties can help increase the net-of-tax return of investors, such as

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multinational firms, and encourage FDI.[1] Despite the expected relationship between tax treaties and FDI, empirical studies have found little evidence to support the relationship. For instance, Blonigen and Davies (2004) use an indicator variable for the presence of a tax treaty and find that the average treaty effect on FDI is insignificant.[2]

Researchers take two different approaches to resolve the puzzling relationship between tax treaties and FDI. The first approach is to use firm-level data to overcome the aggregate nature of FDI data.[3] Davies et al. (2009) use data on the foreign activities of Swedish multinational firms and find positive treaty effects on the probability of FDI into a country. However, they find no significant treaty effect on the level of FDI.

The second approach is to use network optimization techniques to understand the network effect of tax treaties on FDI. By using the Floyd–Warshall algorithm to find shortest paths (i.e., tax-minimizing routes) between countries in a tax treaty network, van't Riet and Lejour (2017) compute tax rates reduced by treaty shopping and centrality measures for conduit countries. These computed variables turn out to be significantly related to FDI.

In line with the "network" approach, I examine the structure of tax-minimizing investment routes in a network of tax treaties between 70 countries. This tax treaty network is represented as a tax rate matrix where each entry is defined as the minimum withholding tax (WHT) rate on dividends and determined by relevant tax treaties and tax laws. In this tax treaty network, investors can minimize tax on dividends by choosing direct investment routes to 26 countries. These countries impose no WHT on dividends or have an imputation system under which no WHT may be imposed on dividends. However, investors may be able to minimize tax on dividends by choosing indirect investment routes to the rest of the 44 countries.

The concept of treaty shopping refers to the use of indirect investment routes through countries with favorable tax treaties. Multinational investors can return after-tax profits to their home countries through tax-minimizing indirect routes. OECD (2015) highlights that treaty shopping is one of the most important sources of concerns regarding the Base Erosion and Profit Shifting (BEPS) project.

Here, I develop an algorithm to analyze the structure of tax-minimizing (direct or indirect) investment routes to the 44 countries. My algorithm works under the

[1] Foreign "direct" investment from country i to country j requires that an investor in country i should play an active role by owning a significant proportion (e.g., at least 10%) of shares of a company in country j. Generally, FDI is distinguished from foreign "portfolio" investment that requires no minimum threshold on ownership. However, FDI can be made with indirect ownership structures using conduit companies in countries other than i and j. In this paper, the definition of indirect investment routes focuses on such indirect ownership structures.

[2] Louie and Rousslang (2008) also use a treaty dummy variable and find no significant treaty effect on the rate of return from FDI after controlling for the quality of governance in source countries. However, di Giovanni (2005) finds a positive treaty effect on cross-border mergers and acquisitions. For a survey of earlier studies, see Davies (2004).

[3] Weyzig (2013) uses Dutch data on special purpose entities and finds positive treaty effects on FDI. Blonigen et al. (2014) use the US BEA data on multinational company operations and find differential effects of tax treaties on foreign affiliate activities, depending on the use of differentiated inputs across industry sectors. They find positive treaty effects on foreign affiliate activities by firms in industries using more differentiated inputs.

constraint that every route passes through at most two countries. I consider two cases about CIT rates in pass-through countries. For brevity, I summarize the results in case (*R*), where pass-through countries impose CIT at reduced rates.

The treaty shopping rate, defined as the difference between the foreign tax rates of the direct route and a tax-minimizing route, is about 3.66 percentage points on average. In relative terms, the treaty shopping rate is about 29.07% of the WHT rate for the direct route. In other words, by using a tax-minimizing route, a multinational investor can reduce the tax rate on dividends by 3.66 percentage points or, in relative terms, can reduce 29.07% of the tax rate. Thus, the use of a tax-minimizing indirect route can substantially reduce tax on dividends incurred by a multinational investor.

For 2954 pairs (61.2%) of residence and source countries, there is a tax-minimizing direct route.[4] For 1876 pairs (38.8%) with no tax-minimizing direct route, on average, the treaty shopping rate is about 9.42 percentage points. In relative terms, the treaty shopping rate is about 74.83% of the WHT rate.

Interestingly, for only 152 pairs (3.1%) of residence and source countries, there are tax-minimizing indirect routes with two pass-through countries. For the rest of country pairs (96.9%), even if it is possible for a route to pass through two countries, every tax-minimizing route passes through at most one country. This is probably because some countries play a central role as a unique pass-through country in a tax-minimizing indirect route. By using the pass-through rate, defined as the relative frequency of being a pass-through country in tax-minimizing indirect routes, I find that the United Kingdom (GB), Malta (MT), Estonia (EE), Slovakia (SK), Hungary (HU), and Cyprus (CY) are among the most common pass-through countries.

In addition, I present examples of tax-minimizing routes with visual diagrams. Notably, from the United States to China, there are tax-minimizing indirect routes passing through Hong Kong or the Netherlands. This finding coincides with anecdotal evidence that American multinational firms, such as Starbucks and IBM, indirectly owned and operated Chinese subsidiaries through foreign equity-holding companies in Hong Kong and the Netherlands.

Multinational firms can choose investment (ownership) structures that influence FDI between countries at an aggregate level. If such investors consider tax as an important factor when choosing investment structures, the results of the network analysis will be consistent with FDI data. I investigate whether the network results help explain FDI patterns by using FDI data from the United Nations Conference on Trade and Development (UNCTAD) and by using standard estimation methods. I also use dummy variables for residence and source countries to control for country-specific factors.

In all relevant regressions, the existence of a tax-minimizing direct route is positively and significantly related to FDI. After controlling for country dummies, I find that the inward FDI stock via a tax-minimizing direct route is about 2.14 times larger than the inward FDI stock via a direct route that is not tax-minimizing. Thus, by making a direct route tax-minimizing, countries can encourage FDI via the direct route.

[4] Given 70 countries, there are 4830 ($= 70 \times 69$) pairs of residence and source countries. Here "residence" refers to a place where investors originally own their assets for investment and "source" refers to a place where investors earn income from their investment.

Furthermore, in all relevant regressions, the treaty shopping rate is negatively and significantly related to FDI. Therefore, even if a direct route is not tax-minimizing, by reducing the treaty shopping rate, countries can encourage FDI via the direct route.

In addition, I find some evidence that the pass-through rate is positively related to FDI. Because increased FDI from pass-through countries may result partly from treaty shopping, source countries can use the pass-through rate as an index to monitor treaty shopping.

This paper is closely related to van't Riet and Lejour (2017). While van't Riet and Lejour (2017) use the Floyd–Warshall algorithm to find tax-minimizing routes between 108 countries, this paper uses a simple computation algorithm for 70 countries. Compared with van't Riet and Lejour (2017), this paper provides an original contribution in three ways. First, this paper proposes new network variables, which turn out to be significantly related to FDI. Second, this paper provides examples and diagrams of tax-minimizing routes, which are consistent with ownership structures of some multinational firms. Third, this paper adopts regression models with country dummies to control for country-specific factors.

From a broader perspective, this paper contributes to the literature on taxation and investment of multinational firms. A cental issue in this literature has been whether and how taxation affects FDI.[5] More recently, researchers focus on how tax systems, together with nontax motives, influence location decisions and ownership structures of multinational firms. Barrios et al. (2012) find that corporate taxation of foreign-source dividends in residence countries may deter entries into potential source countries.[6] Mintz and Weichenrieder (2010) find that German multinational firms use indirect ownership structures through conduit countries and suggest that treaty shopping is a main reason for indirect structures. Lewellen and Robinson (2013) also discover that American multinational firms use indirect ownership structures for tax motives. Dyreng et al. (2015) find that American multinational firms are more likely to use indirect ownership structures (i.e., foreign equity-holding companies) as the WHT rate on dividends paid from a source country to the United States increases. However, they find no evidence that the existence of a tax treaty reduces the use of indirect structures.

The rest of this paper is organized as follows. Section 2 constructs a tax rate matrix to represent a tax treaty network and develops a computation algorithm to find tax-minimizing routes. Section 3 analyzes the structure of tax-minimizing routes in the network and calculates network variables such as the treaty shopping rate. Section 4 studies the relationship between FDI and the network variables. Section 5 concludes.

[5] de Mooij and Ederveen (2003) provide a survey of empirical studies. For recent studies, see Blonigen et al. (2007), Becker et al. (2012a), and Becker and Riedel (2012b). Conversely, Chisik and Davies (2004) study the effects of FDI on tax treaties by using a bargaining model and show that asymmetry in FDI stocks may result in higher tax treaty rates.

[6] For related studies, see Huizinga and Voget (2009) and Voget (2011). Dischinger and Riedel (2011) examine location decisions for intangible assets.

2 Network approach

A complete directed network is a network in which each pair of nodes is connected by two links with opposite directions. Countries i and j can be thought of as nodes i and j in a complete directed network where each link ij is given a weight t_{ij} representing a withholding tax (WHT) rate on dividends. This complete directed network can be thought of as a tax treaty network because tax treaties often determine WHT rates on dividends paid across countries.[7] By constructing a matrix $T = [t_{ij}]$ to represent a tax treaty network, and by developing a computation algorithm, I examine the structure of tax-minimizing investment routes in the tax treaty network.

2.1 Investment routes

An investor plans to invest in country s but lives in country h. From this investment, the investor will earn income m as dividends in country s and repatriate her income to country h. Country s is called the source country, while country h is called the residence (or home) country.

The investor chooses an investment route to maximize her net-of-tax income in the residence country, i.e., to minimize tax when she remits her income to country h. Formally, an investment route, or simply a route, is defined as a series of countries, $h, i, \ldots, j, s$, from country h to country s. Let $R(h, s)$ denote the set of all routes from h to s. Let $r \in R(h, s)$ denote a route. Given a route $r = h, i, \ldots, j, s$, countries i through j are called pass-through countries. A route $h, i, \ldots, j, s$ is often written as

$$h \to i \to \cdots \to j \to s$$

when it is necessary to highlight the direction of the route. The investor can choose an indirect route $h, i, \ldots, j, s$ to invest in country s by establishing conduit entities in countries i through j and making her investment indirectly through these entities. She can also choose the direct route h, s to invest in country s. When the investor remits her income from s to h, the remittance route follows the reverse order of the countries in the investment route. The investor leaves no retained earnings in pass-through countries.

The investor's net-of-tax income depends on tax relief rules in the home country. I assume that the home country has a deduction or an exemption system. The foreign tax rate $f(r)$ of a route r is defined as follows:

$$f(r) = \begin{cases} t_{sh} & \text{if } r = h, s \\ 1 - (1 - t_{sj})(1 - t_j) \cdots (1 - t_i)(1 - t_{ih}) & \text{if } r = h, i, \ldots, j, s \end{cases}$$

Here, t_i denotes the corporate income tax (CIT) rate in country i and t_{ij} denotes the withholding tax (WHT) rate on dividends paid from country i to country j.

If the investor chooses a route r, in the residence country, the investor's net-of-tax income is $m(1 - f(r))(1 - t_h)$. Because the residence country h, the source country

[7] Tax treaties also affect WHT rates on interest and royalties, which may influence cross-border investment in debt instruments and intellectual properties.

s, and the dividend income m are given and fixed, the investor's net-of-tax income is determined by the choice of a route $r \in R(h, s)$. A route $r^* \in R(h, s)$ is called tax-minimizing if for every $r \in R(h, s)$, $f(r^*) \leq f(r)$. The investor can maximize her net-of-tax income by choosing a tax-minimizing route r^*.

Note that the taxation of the residence country is not included in the definitions of foreign tax rate and tax-minimizing route. Because the residence country is given and fixed, if it has a deduction or an exemption system, the taxation of the residence country does not influence the structure of tax-minimizing routes.

Alternatively, if the residence country has a foreign tax credit system, the investor's net-of-tax income is determined by the greater of the foreign tax rate $f(r)$ and the home CIT rate t_h. Thus, if the home CIT rate is sufficiently high under the credit system, the direct route can yield the same net-of-tax income as a tax-minimizing indirect route, and the investor may have no incentive to use an indirect route.[8] However, the investor has such an incentive if the residence country has a deduction or an exemption system.

2.2 Tax rate matrix

A tax rate matrix $T = [t_{ij}]$ shows WHT rates on dividends paid across national borders.[9] Each WHT rate t_{ij} is applied for a pair of countries i and j. If an entity in country i remits dividends to an entity in country j, the tax agency in country i imposes WHT on dividends at the rate t_{ij}. Each WHT rate t_{ij} is determined by the tax treaty between countries i and j and by the national tax law of country i. For the sake of running an algorithm, each diagonal entry t_{ii} is assumed to be zero.

I construct a tax rate matrix $T = [t_{ij}]$ between 70 countries, as listed in Tables 1 and 2, by using the information on tax treaties and tax laws from Deloitte International Tax Source and PwC Worldwide Tax Summaries.[10] From these Web sites, I also find the information on CIT rates. Tables 1 and 2 summarize notable WHT systems with minimum, median, and maximum WHT rates and CIT rates in the 70 countries, as of July 2015.

Each WHT rate t_{ij} is determined as follows:

First, if the European Union Parent-Subsidiary Directive (EU PSD) can be applied for countries i and j, then set $t_{ij} = 0$. The EU PSD requires the exemption of WHT on dividends paid across EU countries, Iceland, Norway, and Switzerland.[11]

Second, if the EU PSD cannot be applied for at least one of countries i and j, then check if there is a tax treaty between countries i and j. If there is no tax treaty between i and j, then set t_{ij} equal to the WHT rate specified by the national tax law of country i.

[8] Notably, until 2017, the United States maintained a foreign tax credit system with a high CIT rate. However, by the Tax Cuts and Jobs Act of 2017, it adopts an exemption system for foreign-source dividends.

[9] Barrios et al. (2012, Table 2) provide a tax rate matrix of 33 European countries. Johannesen (2012, Table 1) also presents a matrix of WHT rates on interest paid across 28 OECD countries.

[10] Accessed in July 2015 at dits.deloitte.com and taxsummaries.pwc.com.

[11] European Commission, Council Directive 2003/123/EC of 22 December 2003.

Table 1 Selected countries and tax systems (2015)

Country	Code	WHT system	Min	Med	Max	CIT
Argentina	AR	Single rate 10%	10	10	10	35
Australia	AU	Imputation	0	0	0	30
Austria	AT		0	5	25	25
Belgium	BE		0	0	25	33
Bermuda	BM	No WHT	0	0	0	0
Brazil	BR	No WHT	0	0	0	34
Bulgaria	BG		0	5	5	10
Canada	CA		5	5	25	15
Cayman	KY	No WHT	0	0	0	0
Chile	CL	Imputation	0	0	0	22.5
China	CN		5	10	10	25
Colombia	CO	Imputation	0	0	0	25
Croatia	HR		0	5	12	20
Cyprus	CY	No WHT	0	0	0	12.5
Czech Republic	CZ		0	5	15	19
Denmark	DK		0	0	27	23.5
Ecuador	EC	Imputation	0	0	0	22
Egypt	EG		0	10	10	22.5
Estonia	EE	No WHT	0	0	0	20
Finland	FI		0	5	20	20
France	FR		0	0	30	33.3
Germany	DE		0	5	25	15
Gibraltar	GI	No WHT	0	0	0	10
Greece	GR		0	5	10	29
Guernsey	GG	No WHT	0	0	0	0
Hong Kong	HK	No WHT	0	0	0	16.5
Hungary	HU	No WHT	0	0	0	10
Iceland	IS		0	5	18	20
India	IN	Single rate 15%	15	15	15	30
Indonesia	ID		5	12.5	20	25
Ireland	IE		0	0	20	12.5
Israel	IL		0	10	25	26.5
Italy	IT		0	5	26	27.5
Japan	JP		0	10	20	23.9
Korea	KR		0	7	20	22

Third, if the EU PSD cannot be applied for at least one of countries i and j, but if there is a tax treaty between countries i and j, then set t_{ij} equal to the minimum of the rates specified by the tax treaty between countries i and j and by the national tax law of country i.

Table 2 Selected countries and tax systems (2015)

Country	Code	WHT system	Min	Med	Max	CIT
Latvia	LV	No WHT	0	0	0	15
Lithuania	LT		0	5	15	15
Luxembourg	LU		0	0	15	21
Malaysia	MY	No WHT	0	0	0	25
Malta	MT	No WHT	0	0	0	35
Mauritius	MU	No WHT	0	0	0	15
Mexico	MX		0	5	15	30
Netherlands	NL		0	0	15	25
New Zealand	NZ	Imputation	0	0	0	28
Norway	NO		0	5	25	27
Panama	PA		0	10	10	25
Peru	PE		5	6.8	6.8	28
Philippines	PH		5	15	15	30
Poland	PL		0	5	19	19
Portugal	PT		0	5	25	21
Romania	RO		0	5	16	16
Russia	RU		5	10	15	20
Saudi Arabia	SA		0	5	5	20
Singapore	SG	No WHT	0	0	0	17
Slovakia	SK	No WHT	0	0	0	22
Slovenia	SI		0	5	15	17
South Africa	ZA		5	5	15	28
Spain	ES		0	0	20	28
Sweden	SE		0	0	30	22
Switzerland	CH		0	0	35	8.5
Taiwan	TW		5	20	20	17
Thailand	TH		5	10	10	20
Turkey	TR		5	10	15	20
Ukraine	UA		0	5	15	18
Arab Emirates	AE	No WHT	0	0	0	0
United Kingdom	GB	No WHT	0	0	0	20
United States	US		0	5	30	35
Uruguay	UY		5	7	7	25
Venezuela	VE	Imputation	0	0	0	34
Vietnam	VN	No WHT	0	0	0	22

A tax treaty may specify a system of WHT rates on dividends depending on the percentage of shares. Given a system of WHT rates, I assume that the minimum WHT rate is applied. For example, according to the tax treaty between China (CN) and the United Kingdom (GB), if a Chinese entity pays dividends to a British entity that holds

at least 25% of its shares, the WHT rate is 5%. Otherwise, the WHT rate is 10%. In this case, the minimum WHT rate of 5% is applied, i.e., for $i = \text{CN}$ and $j = \text{GB}$, $t_{ij} = 5$.

Moreover, a tax treaty may include the limitation on benefits (LOB) clause, specifying additional residency requirements to obtain the benefits of the tax treaty. If the LOB clause specifies that the minimum WHT rate can only be applied to certain pension funds, I take the next lowest WHT rate for the matrix. Otherwise, I assume that the minimum WHT rate is applied. For instance, according to the tax treaty (amended and signed in 2010) between the United States (US) and Hungary (HU), the minimum WHT rate is 0%, but this rate can only be applied to pension funds qualified by the LOB clause of the tax treaty. The next lowest rate is 5%, which can be applied to other corporate entities. So, for $i = \text{US}$ and $j = \text{HU}$, it is assumed that $t_{ij} = 5$. However, according to the tax treaty between the United States (US) and Japan (JP), the minimum WHT rate is 0%, and this rate can be applied to pension funds as well as companies holding more than 50% of shares for a period longer than 12 months. So, for $i = \text{US}$ and $j = \text{JP}$, it is assumed that $t_{ij} = 0$.

In addition to the general rules explained so far, I consider the following specific cases to construct the tax rate matrix.

Subnational (state, province, etc.) governments may also impose WHT on dividends. However, it is an onerous task to find out the WHT systems of all the subnational governments in 70 countries. Even if a subnational government imposes WHT on dividends, the rate is usually very low. Therefore, I only consider WHT imposed by national (federal) governments.

There is no WHT on dividends in India. However, India imposes a dividend distribution tax at the rate of 15%. Hence, it is realistic to assume that the WHT rate is 15% in India.

Australia and New Zealand have dividend imputation systems under which no WHT may be imposed on dividends distributed by companies that already pay CIT. Chile, Colombia, Ecuador, and Venezuela also have such systems. I assume that these countries impose no WHT on dividends.

Belgium imposes no WHT on dividends paid to entities resident in a country with a tax treaty. Ireland has a similar rule for WHT exemption. In Denmark, the WHT rate is no greater than 15% if dividends are paid to shareholders resident in a country with a tax treaty.

In the tax rate matrix as constructed above, even without using a computation algorithm, we can find that the investor can minimize tax on dividends by investing directly to 26 countries among all the countries in Tables 1 and 2. In particular, 18 countries impose no WHT on dividends and six countries have a dividend imputation system. No WHT may be imposed on dividends under a dividend imputation system. Also, two countries have a WHT system with a single rate for all countries.

However, the investor may be able to minimize tax on dividends by investing indirectly to the rest of the 44 countries. With the information about tax rates at hand, the investor needs to develop a computation algorithm to find tax-minimizing investment routes to these 44 countries.

2.3 Algorithm

Given a tax rate matrix $T = [t_{ij}]$ and a pair (h, s) of residence and source countries, the investor can find tax-minimizing routes from country h to country s by using a computation algorithm. I propose a computation algorithm that works under a constraint on the number of pass-through countries. This algorithm repeats the following three steps for all pairs of residence and source countries. Let (h, s) be a pair of countries.

Step 1. Build a list of all routes from country h to country s under the constraint that the number of pass-through countries in each route is at most two and compute the net-of-tax income for each route.

Step 2. Find the maximum net-of-tax income among all routes in the list.

Step 3. Find and list all the routes from h to s that achieve the maximum net-of-tax income.

Given 70 countries, there are 4830 ($= 70 \times 69$) pairs of residence and source countries. Because there are at most two pass-through countries in a route, the list of all routes includes an indirect route $h \to i \to j \to s$ with two pass-through countries i and j, an indirect route $h \to j \to s$ with a pass-through country j, and the direct route $h \to s$. Given 70 countries, for each pair of residence and source countries, there are 4625 ($= 68 \times 67 + 68 + 1$) routes in the list of Step 1.

As the maximum number of pass-through countries in a route increases, the number of routes in the list of Step 1 grows exponentially, and the computational burden (or running time) of the algorithm increases exponentially. To limit the computational burden, I assume that there are at most two pass-through countries in a route.[12]

3 Network results

For an indirect route, the net-of-tax income depends on corporate income tax (CIT) rates in pass-through countries. I consider two cases about CIT rates. In the first case (R), every pass-through country imposes CIT at a reduced rate of 1%.[13] In the second case (S), every pass-through country imposes CIT at the statutory rate. One can expect that tax-minimizing indirect routes tend to pass through countries with no WHT in case (R), and pass through countries with no CIT in case (S).

To present the results, I introduce the following network variables. For a pair of countries i and j, if there is a tax-minimizing direct route $i \to j$ in case (R),

[12] This assumption may not be unrealistic. Mintz and Weichenrieder (2010, Table 4) find that only 0.2% of German multinational firms use cross-border investment routes (ownership chains) with three or more pass-through countries.

[13] In a pass-through country, CIT can be reduced by a favorable tax system for conduit entities, such as foreign equity-holding companies. According to van't Riet and Lejour (2017, Table 4), the average effective CIT rate in pass-through (or conduit) countries is about 0.3%. In case (R), every pass-through country is assumed to impose CIT at the reduced rate of 1%. This rate is chosen as the smallest positive integer, and it is also greater than the estimate of 0.3%. As far as the reduced CIT rate is small and positive, the structure of tax-minimizing routes presented in this section remains qualitatively unchanged. However, in a special case when the reduced CIT rate is zero, tax-minimizing routes tend to pass through more countries with no WHT, because there is no additional cost of adding such a pass-through country into an indirect route. This special case is less realistic and not reported in this section.

Table 3 Descriptive statistics for network variables

Variable	Description	Obs	Mean	SD	Min	Max
DIRECT_{ij}	Existence of a tax-minimizing direct route from i to j	4830	0.61	0.49	0	1
DIRECTS_{ij}		4830	0.75	0.43	0	1
TSHOP_{ij}	Treaty shopping rate among routes from i to j	4830	3.66	6.27	0	34
TSHOPS_{ij}		4830	2.29	5.03	0	30
PASS_{ij}	Percentage of tax-minimizing indirect routes to j through i	4830	0.98	4.15	0	100
PASSS_{ij}		4830	2.51	11.58	0	100

$\text{DIRECT}_{ij} = 1$, and otherwise, $\text{DIRECT}_{ij} = 0$. Similarly, in case (S), let DIRECTS_{ij} denote the indicator variable for the existence of a tax-minimizing direct route $i \to j$.

For a pair of countries i and j, the treaty shopping rate among routes from i to j in case (R) is defined as $\text{TSHOP}_{ij} = t_{ji} - f(r^*)$, where r^* is a tax-minimizing route from i to j in case (R). Similarly, let TSHOPS_{ij} denote the treaty shopping rate among routes from i to j in case (S).

In words, the treaty shopping rate among routes between a pair of residence and source countries is defined as the difference between the foreign tax rates of the direct route and a tax-minimizing route.[14] An investor can reduce her foreign tax rate by the treaty shopping rate by using a tax-minimizing route. If the direct route is tax-minimizing, the treaty shopping rate is zero. If not, the treaty shopping rate is a positive number in percentage points.

For a pair of countries i and j, the pass-through rate of i to j in case (R), denoted by PASS_{ij}, is defined as the percentage of tax-minimizing indirect routes to j passing through i among all tax-minimizing indirect routes to j in case (R). If there is no tax-minimizing indirect route to j in case (R), let $\text{PASS}_{ij} = 0$. Similarly, let PASSS_{ij} denote the pass-through rate of i to j in case (S).

The pass-through rate of i to j shows the relative frequency of i being a pass-through country in tax-minimizing indirect routes to j. The pass-through rate is comparable to the notion of betweenness centrality in van't Riet and Lejour (2017). To see which countries are common pass-through countries in tax-minimizing indirect routes, I will rank the countries according to their pass-through rates and compare the rankings of pass-through (or conduit) countries with those in van't Riet and Lejour (2017).

Table 3 presents descriptive statistics for network variables. Note that DIRECT_{ij} and TSHOP_{ij} are defined for a pair of residence country i and source country j whereas PASS_{ij} is defined for a pair of pass-through country i and source country j.

Table 3 shows that direct routes are tax-minimizing for more than the majority of country pairs. In case (R), for 2954 pairs (61.2%) of countries i and j, the direct route $i \to j$ is tax-minimizing. In case (S), for 3637 pairs (75.3%) of countries i and j, the direct route $i \to j$ is tax-minimizing.

[14] Note that the taxation of the residence country is not included in the definitions of foreign tax rate and tax-minimizing route. The absence of the residence country taxation may underestimate the magnitude of treaty shopping.

Table 3 also shows that the use of a tax-minimizing indirect route can substantially reduce tax on dividends incurred by a multinational investor. In case (R), on average, the treaty shopping rate is about 3.66 percentage points. In relative terms, the treaty shopping rate is about 29.07% of the WHT rate for the direct route. By using a tax-minimizing route, a multinational investor can reduce the tax rate on dividends by 3.66 percentage points or, in relative terms, can reduce 29.07% of the tax rate.

The treaty shopping rate is more substantial when the average is taken over the pairs of countries with no tax-minimizing direct route. In case (R), for 1876 pairs (38.8%) of countries i and j, the direct route $i \to j$ is not tax-minimizing. Among these 1876 pairs with no tax-minimizing direct route, on average, the treaty shopping rate is about 9.42 percentage points. In relative terms, the treaty shopping rate is about 74.83% of the WHT rate. In case (S), for 1193 pairs (24.7%) of countries i and j, the direct route $i \to j$ is not tax-minimizing. Among these 1193 pairs, on average, the treaty shopping rate is about 9.29 percentage points. In relative terms, the treaty shopping rate is about 63.23% of the WHT rate.

3.1 Reduced rates in pass-through countries

As mentioned in Sect. 2.2, investors can minimize tax on dividends by investing directly to 26 countries (with no WHT, dividend imputation, or single-rate systems) among the 70 countries. However, investors may be able to minimize tax on dividends by investing indirectly to the rest of the 44 source countries.

In case (R), when pass-through countries impose CIT at the reduced rate, for 1876 pairs of countries i and j, the direct route $i \to j$ is not tax-minimizing. For a typical pair of countries with no tax-minimizing direct route, multiple tax-minimizing indirect routes exist. Mostly, such indirect routes pass through only one country. For 1724 pairs, out of the 1876 pairs with no tax-minimizing direct route, there are tax-minimizing indirect routes with one pass-through country. For 152 pairs, there are tax-minimizing indirect routes with two pass-through countries.

Precisely, when Korea (KR), Peru (PE), the Philippines (PH), Saudi Arabia (SA), or Ukraine (UA) is the source country, there are tax-minimizing indirect routes with two pass-through countries. However, for the rest of the countries, even if it is possible for a route to pass through two countries, every tax-minimizing route passes through at most one country.

Table 4 shows the types and numbers of tax-minimizing routes to the 44 source countries in case (R). Column "Direct route" shows the number of tax-minimizing direct routes to each source country.[15] Because there is only one direct route between a pair of residence and source countries, this number coincides with the number of residence countries with tax-minimizing direct routes to each source country. Column "1 Indirect route" shows the number of tax-minimizing indirect routes with one pass-through country, and column "1 Indirect home" shows the number of residence countries with such routes. Column "2 Indirect route" shows the number of tax-minimizing indirect routes with two pass-through countries and column "2 Indirect home" shows the num-

[15] For country j, the number in column "Direct route" is calculated as $\sum_{i \neq j} \text{DIRECT}_{ij}$.

ber of residence countries with such routes. Column "TSHOP" shows the average of treaty shopping rates for each source country.[16]

For example, as Table 4 shows, when Austria (AT) is the source country, there are 33 tax-minimizing direct routes from 33 residence countries to Austria. Moreover, there are 464 tax-minimizing indirect routes from 36 residence countries to Austria. There are no tax-minimizing indirect routes with two pass-through countries. Thus, every tax-minimizing indirect route to Austria passes though only one country. When Austria is the source country, the average treaty shopping rate is about 6.83 percentage points.

Table 4 also shows that Belgium (BE), Ireland (IE), and Thailand (TH) have the largest numbers of tax-minimizing direct routes. There are 61 tax-minimizing direct routes to Belgium, 54 to Ireland, and 69 to Thailand. In contrast, Indonesia (ID), Korea (KR), Panama (PA), Peru (PE), and the Philippines (PH) have the smallest number of tax-minimizing direct routes. There is only 1 tax-minimizing direct route to each of these countries.

The structure of tax-minimizing routes can be examined from the perspective of a residence country. Table 5 shows the types and numbers of tax-minimizing routes from capital-exporting residence countries in case (R).[17] Column "Direct route" shows the number of tax-minimizing direct routes from each residence country.[18] This number coincides with the number of source countries with tax-minimizing direct routes from each residence country. Column "1 Indirect route" shows the number of tax-minimizing indirect routes with 1 pass-through country, and column "1 Indirect source" shows the number of source countries with such routes. Column "2 Indirect route" shows the number of tax-minimizing indirect routes with two pass-through countries, and column "2 Indirect source" shows the number of source countries with such routes. Column "TSHOP" shows the average of treaty shopping rates for each residence country.[19]

For instance, as Table 5 shows, when Austria (AT) is the residence country, there are 55 tax-minimizing direct routes from Austria to 55 source countries. Moreover, there are 90 tax-minimizing indirect routes with 1 pass-through country from Austria to 13 source countries. There are 16 tax-minimizing indirect routes with two pass-through countries from Austria to a source country, namely Korea. When Austria is the residence country, the average treaty shopping rate is about 1.18 percentage points.

Among the capital-exporting residence countries listed in Table 5, France (FR), Sweden (SE), and Switzerland (CH) have the largest numbers of tax-minimizing direct routes. There are 61 tax-minimizing direct routes from France, 60 from Sweden, and 60 from Switzerland. In contrast, Canada (CA), Japan (JP), and Korea (KR) have the smallest numbers of tax-minimizing direct routes. There are 32 tax-minimizing direct routes from Canada, 37 from Japan, and 35 from Korea.

[16] For country j, the number in column "TSHOP" is $\sum_{i \neq j} \mathrm{TSHOP}_{ij}/(70-1)$.

[17] Each of the 70 countries, as a residence country, has tax-minimizing indirect routes to some source countries. Table 5 shows the results for the 20 countries whose outward FDI stock is greater than the inward stock according to the FDI data of the UNCTAD.

[18] For country i, the number in column "Direct route" is calculated as $\sum_{j \neq i} \mathrm{DIRECT}_{ij}$.

[19] For country i, the number in column "TSHOP" is $\sum_{j \neq i} \mathrm{TSHOP}_{ij}/(70-1)$.

Table 4 Tax-minimizing routes to source countries in case (*R*)

Source	Direct route	1 Indirect route	1 Indirect home	2 Indirect route	2 Indirect home	TSHOP
AT	33	464	36	0	0	6.83
BE	61	162	8	0	0	2.78
BG	30	398	39	0	0	2.26
CA	40	494	29	0	0	4.46
CN	31	590	38	0	0	2.11
HR	31	426	38	0	0	4.99
CZ	31	387	38	0	0	4.81
DK	39	410	30	0	0	7.41
EG	5	232	64	0	0	7.84
FI	34	415	35	0	0	6.01
FR	38	391	31	0	0	5.78
DE	31	386	38	0	0	6.62
GR	30	398	39	0	0	4.51
IS	30	398	39	0	0	8.06
ID	1	68	68	0	0	8.24
IE	54	245	15	0	0	4.13
IL	5	220	64	0	0	12.71
IT	31	386	38	0	0	7.35
JP	9	318	60	0	0	10.62
KR	1	18	18	610	50	7.38
LT	32	411	37	0	0	6.13
LU	35	438	34	0	0	4.14
MX	19	610	50	0	0	5.04
NL	40	391	29	0	0	4.07
NO	33	433	36	0	0	7.59
PA	1	68	68	0	0	8.28
PE	1	38	38	445	30	0.63
PH	1	31	31	378	37	7.04
PL	31	421	38	0	0	6.36
PT	30	398	39	0	0	8.06
RO	31	421	38	0	0	6.41
RU	27	562	42	0	0	3.91
SA	3	121	53	514	13	3.64
SI	30	398	39	0	0	6.29
ZA	40	528	29	0	0	3.30
ES	38	457	31	0	0	4.75
SE	36	391	33	0	0	7.49

Table 4 continued

Source	Direct route	1 Indirect route	1 Indirect home	2 Indirect route	2 Indirect home	TSHOP
CH	36	458	33	0	0	6.36
TW	17	218	52	0	0	6.60
TH	69	0	0	0	0	0.00
TR	15	446	54	0	0	5.20
UA	5	160	42	1241	22	7.19
US	12	393	57	0	0	11.97
UY	13	350	56	0	0	0.85

Table 5 Tax-minimizing routes from home countries in case (R)

Home	Direct route	1 Indirect route	1 Indirect source	2 Indirect route	2 Indirect source	TSHOP
AT	55	90	13	16	1	1.18
CA	32	308	35	20	2	2.76
DK	57	74	12	0	0	1.05
FI	59	28	10	0	0	0.98
FR	61	36	8	0	0	0.53
DE	59	46	9	17	1	0.74
GR	55	73	13	16	1	1.88
IS	53	121	15	16	1	1.92
IE	55	86	13	16	1	1.23
IT	53	103	15	16	1	1.56
JP	37	323	30	32	2	3.56
KR	35	264	31	85	3	3.15
LU	57	62	11	16	1	0.90
NL	59	53	9	18	1	0.63
NO	54	102	15	0	0	1.73
ES	57	78	11	16	1	1.25
SE	60	37	9	0	0	0.66
CH	60	30	9	0	0	0.55
GB	58	69	11	0	0	0.92
US	41	316	27	21	1	2.55

Table 6 shows the most common pass-through countries in case (R). Column "Treaty" shows the number of tax treaties that a country has with any one of the 70 countries. Column "PASS" shows the total of pass-through rates for each pass-through country.[20] Column "Rank" shows the ranking of a pass-through country based on the

[20] For country i, the number in column "PASS" is $\sum_{j \neq i} \text{PASS}_{ij}$.

Table 6 Common pass-through countries in case (*R*)

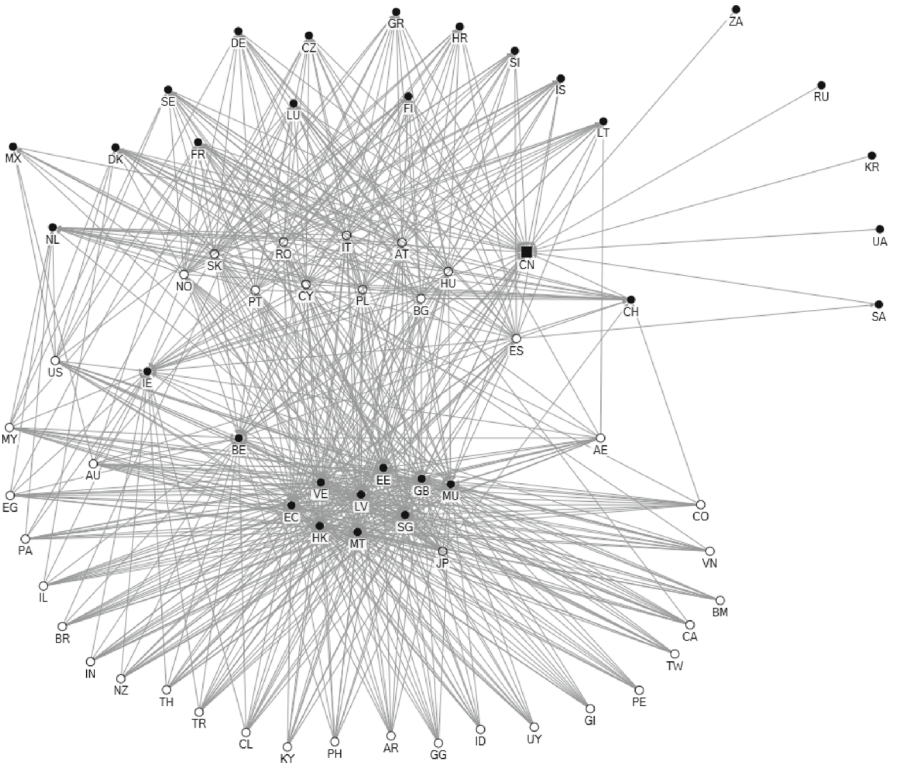
Country	WHT system	Treaty	PASS	Rank	vRL rank
GB	No WHT	60	411.41	1	1
MT	No WHT	51	312.81	2	10
EE	No WHT	43	297.18	3	3
SK	No WHT	49	283.71	4	8
HU	No WHT	56	255.44	5	5
LV	No WHT	41	251.93	6	31
CY	No WHT	39	245.12	7	9
BE		61	225.09	8	17
HK	No WHT	27	195.40	9	22
ES		62	194.69	10	15

total pass-through rate. For comparison, column "vRL Rank" shows the ranking of a pass-through (or conduit) country in van't Riet and Lejour (2017).

As shown in Table 6, the United Kingdom (GB), Malta (MT), Estonia (EE), Slovakia (SK), Hungary (HU), and Cyprus (CY) are ranked in the 10 most common pass-through countries by the total pass-through rate as well as by the betweenness centrality in van't Riet and Lejour (2017). As of July 2015, all these countries impose no WHT on dividends and belong to the European Union.[21] Latvia (LV), also ranked in the top 10 by the total pass-through rate, has the same features of WHT exemption and EU membership. In addition, Belgium (BE), Hong Kong (HK), and Spain (ES) are among the most common pass-through countries. While generally imposing WHT on dividends, Belgium and Spain have the largest numbers of tax treaties within the 70 countries and keep the WHT rates close to zero. Hong Kong is ranked highest among the non-EU countries. Outside the top ten countries, Ireland (11th), the Netherlands (17th), and Singapore (19th) are ranked high.

In Appendix A, I present examples of tax-minimizing routes between selected pairs of residence and source countries. As in Table 15, when the United States (US) is the residence country and China (CN) is the source country, there are 18 tax-minimizing indirect routes that pass through either one of the following countries: Belgium (BE), Denmark (DK), Ecuador (EC), Estonia (EE), Finland (FI), France (FR), Germany (DE), Hong Kong (HK), Ireland (IE), Latvia (LV), Malta (MT), Mauritius (MU), Mexico (MX), the Netherlands (NL), Singapore (SG), Sweden (SE), the United Kingdom (GB), and Venezuela (VE). By definition, these tax-minimizing indirect routes must yield the same net-of-tax income. Table 15 shows that the treaty shopping rate among routes between the United States and China is 4.05 percentage points.

[21] Tax-minimizing indirect routes tend to pass through countries with WHT exemption and EU membership because entities in such countries can receive WHT-exempt dividends from other EU countries under the EU PSD, as mentioned in Sect. 2.2, and pay WHT-exempt dividends to any other countries under the WHT exemption rule.



Note: ■ CN source; ● countries with tax-minimizing direct routes; ○ countries with no such routes

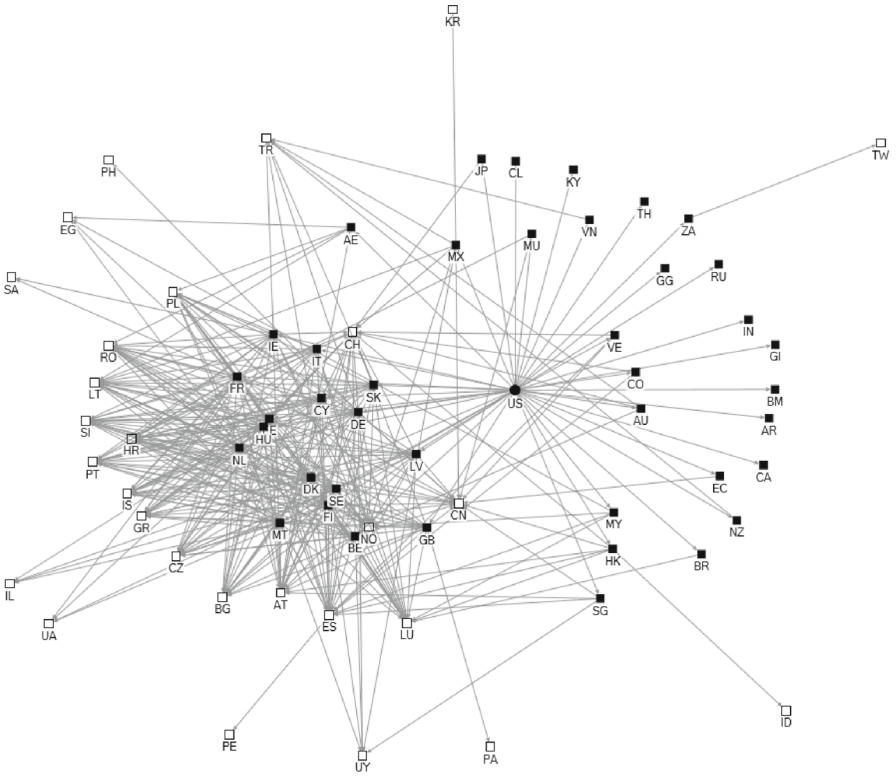
Fig. 1 Tax-minimizing routes to China (CN)

Interestingly, from the United States (US) to China (CN), there are tax-minimizing indirect routes that pass through Hong Kong (HK) or the Netherlands (NL) as follows:

$$\text{US} \rightarrow \text{HK} \rightarrow \text{CN}$$
$$\text{US} \rightarrow \text{NL} \rightarrow \text{CN}$$

This finding is consistent with anecdotal evidence that Starbucks Corporation, an American multinational firm, indirectly owned and operated Starbucks (China) Company Ltd., a Chinese subsidiary, through Starbucks Asia Pacific Investment Holding II Ltd., a holding company in Hong Kong. IBM Corporation also indirectly owned and operated IBM China Company Ltd., a Chinese subsidiary, through IBM Holdings B.V., a Dutch holding company.

The structure of tax-minimizing routes can be examined visually with network diagrams. To draw the network diagrams, I use NodeXL with the Harel–Koren layout, which places closely connected nodes (i.e., countries) in proximal locations.



Note: ● US residence; ■ countries with tax-minimizing direct routes; □ countries with no such routes

Fig. 2 Tax-minimizing routes from the United States (US)

Figure 1 shows a network diagram of tax-minimizing routes to China (CN). Here, the solid square stands for the source country. Each solid circle stands for a residence country with a tax-minimizing direct route. Each empty circle stands for a residence country with a tax-minimizing indirect route. Each arrow shows the direction of a route.

As shown in the row CN of Table 4 and in Fig. 1, there are 31 tax-minimizing direct routes from 31 residence countries to China. Moreover, there are 590 tax-minimizing indirect routes from 38 residence countries to China. There are no tax-minimizing indirect routes with two pass-through countries.

Figure 2 shows a network diagram of tax-minimizing routes from the United States (US). Here, the solid circle stands for the residence country. Each solid square stands for a source country with a tax-minimizing direct route. Each empty square stands for a source country with a tax-minimizing indirect route. Each arrow shows the direction of a route.

As shown in the row US of Table 5 and in Fig. 2, there are 41 tax-minimizing direct routes from the United States to 41 source countries. Moreover, there are 316 tax-minimizing indirect routes with one pass-through country from the United States to 27 source countries. There are 21 tax-minimizing indirect routes with two pass-through countries from the United States to a source country, namely, Peru.

In Appendix B, Figs. 3, 4, 5 and 6 show network diagrams of tax-minimizing investment routes to Mexico, Indonesia, Poland, and Saudi Arabia, respectively, in case (*R*). These four countries and China are the countries with the largest net inward FDI stock according to the FDI data of the UNCTAD.

Figures 7, 8, 9 and 10 show network diagrams of tax-minimizing investment routes from Japan, France, Germany, and the United Kingdom, respectively, in case (*R*). These four countries and the United States are the countries with the largest net outward FDI stock.

3.2 Statutory rates in pass-through countries

In case (*S*), when pass-through countries impose CIT at the statutory rates, for 1193 pairs of countries i and j, the direct route $i \to j$ is not tax-minimizing.

In case (*S*), all tax-minimizing indirect routes pass through only the following six countries: Bermuda (BM), Cayman (KY), Guernsey (GG), Hungary (HU), Switzerland (CH), and the United Arab Emirates (AE). All these countries except for Hungary and Switzerland impose no CIT. Hungary and Switzerland impose CIT at relatively low statutory rates, 10 and 8.5%, respectively.

In case (*S*), because Bermuda, Cayman, Guernsey, and the United Arab Emirates impose no CIT and no WHT, by adding one of these countries into a tax-minimizing indirect route with one pass-through country, we can construct a tax-minimizing indirect route with two pass-through countries. Hence, there are more tax-minimizing indirect routes with two pass-through countries than those with one pass-through country. However, this sort of extension does not change the net-of-tax income.

Table 7 shows the types and numbers of tax-minimizing routes to the 44 source countries in case (*S*). The columns are defined similarly as in Table 4. For example, in Table 7, when Austria (AT) is the source country, there are 33 tax-minimizing direct routes from 33 residence countries to Austria. Moreover, there are 68 tax-minimizing indirect routes with one pass-through country from 68 residence countries to Austria. Also, there are 201 tax-minimizing indirect routes with two pass-through countries from 68 residence countries to Austria. When Austria is the source country, the average treaty shopping rate is about 7.35 percentage points.

Table 7 also shows that Bulgaria (BG), Peru (PE), Saudi Arabia (SA), Thailand (TH), and Uruguay (UY) have the largest number of tax-minimizing direct routes. There are 69 tax-minimizing direct routes to each of these countries. In contrast, Egypt (EG), Mexico (MX), and Panama (PA) have the smallest numbers of tax-minimizing direct routes. There are five tax-minimizing direct routes to Egypt, 19 to Mexico, and nine to Panama.

Table 8 shows the types and numbers of tax-minimizing routes from capital-exporting residence countries in case (*S*). The columns are defined similarly as in

Table 7 Tax-minimizing routes to source countries in case (*S*)

Source	Direct route	1 Indirect route	1 Indirect home	2 Indirect route	2 Indirect home	TSHOP
AT	33	68	68	201	68	7.35
BE	61	68	68	201	68	2.90
BG	69	152	39	444	39	0.00
CA	40	68	68	201	68	4.86
CN	33	37	37	108	37	1.54
HR	43	28	28	84	22	0.88
CZ	46	37	37	108	37	2.61
DK	48	27	27	88	23	3.80
EG	5	68	68	201	68	8.77
FI	34	68	68	201	68	6.52
FR	38	68	68	201	68	6.23
DE	48	37	37	108	37	4.42
GR	38	38	38	111	38	2.25
IS	39	32	32	104	27	3.59
ID	34	67	67	201	68	4.25
IE	54	68	68	201	68	4.35
IL	39	30	30	72	19	4.07
IT	40	37	37	108	37	5.14
JP	21	59	59	177	60	7.14
KR	34	67	67	198	67	4.16
LT	32	68	68	201	68	6.67
LU	52	66	34	192	34	2.17
MX	19	68	68	201	68	5.77
NL	50	28	28	81	28	2.39
NO	47	26	26	92	24	3.67
PA	9	66	66	195	66	4.35
PE	69	268	68	792	68	0.00
PH	20	67	67	198	67	3.55
PL	31	68	68	201	68	6.91
PT	37	38	38	111	38	5.80
RO	31	68	68	201	68	6.96
RU	64	61	20	156	15	0.14
SA	69	260	66	768	66	0.00
SI	39	38	38	111	38	4.03
ZA	47	22	22	52	14	0.32
ES	46	30	30	90	31	2.96
SE	52	24	24	80	21	3.91
CH	50	32	32	96	33	4.45
TW	21	48	48	92	24	1.17

Table 7 continued

Source	Direct route	1 Indirect route	1 Indirect home	2 Indirect route	2 Indirect home	TSHOP
TH	69	268	68	792	68	0.00
TR	41	53	53	159	54	2.03
UA	38	63	63	189	64	3.80
US	44	25	25	76	20	4.78
UY	69	220	56	648	56	0.00

Table 8 Tax-minimizing routes from home countries in case (*S*)

Home	Direct route	1 Indirect route	1 Indirect source	2 Indirect route	2 Indirect source	TSHOP
AT	64	134	48	387	47	0.54
CA	54	148	59	434	59	1.15
DK	64	134	48	390	48	0.51
FI	63	128	45	369	44	0.47
FR	67	127	44	369	44	0.14
DE	65	128	45	372	45	0.36
GR	60	136	50	390	48	1.15
IS	59	137	51	390	48	1.03
IE	62	129	46	372	45	0.54
IT	60	136	50	393	49	0.80
JP	49	152	62	429	57	2.05
KR	52	148	61	436	61	1.42
LU	65	129	46	375	46	0.41
NL	66	132	46	384	46	0.22
NO	60	137	51	390	48	0.79
ES	62	123	46	354	45	0.57
SE	65	132	46	384	46	0.29
CH	68	128	45	372	45	0.07
GB	65	129	46	372	45	0.32
US	54	149	59	438	59	1.10

Table 5. For instance, in Table 8, when Austria (AT) is the residence country, there are 64 tax-minimizing direct routes from Austria to 64 source countries. Moreover, there are 134 tax-minimizing indirect routes with one pass-through country from Austria to 48 source countries. Also, there are 387 tax-minimizing indirect routes with two pass-through countries from Austria to 47 source countries. When Austria is the residence country, the average treaty shopping rate is about 0.54 percentage points.

Among the capital-exporting residence countries listed in Table 8, France (FR), the Netherlands (NL), and Switzerland (CH) have the largest numbers of tax-minimizing

Table 9 Common pass-through countries in case (*S*)

Country	WHT system	Treaty	PASS	Rank	vRL rank
AE	No WHT	45	4397.98	1	19
GG	No WHT	7	2294.41	2	46
BM	No WHT	0	2256.82	3	44
KY	No WHT	0	2256.82	3	45
HU	No WHT	56	839.22	5	5
CH		61	63.08	6	18

direct routes. There are 67 tax-minimizing direct routes from France, 66 from the Netherlands, and 68 from Switzerland. In contrast, Canada (CA), Japan (JP), Korea (KR), and the United States (US) have the smallest numbers of tax-minimizing direct routes. There are 54 tax-minimizing direct routes from Canada, 49 from Japan, 52 from Korea, and 54 from the United States.

Table 9 shows the rankings of the six pass-through countries in case (*S*). The columns are defined similarly as in Table 6. In case (*S*), all tax-minimizing indirect routes pass through only the 6 countries listed in Table 9. Except for Hungary (HU), the rankings in case (*S*) turn out to be very different from those in case (*R*) and in van't Riet and Lejour (2017). Bermuda (BM), Cayman (KY), Guernsey (GG), and the United Arab Emirates (AE) are ranked high in case (*S*). These countries impose no CIT and no WHT on dividends. Moreover, except for the United Arab Emirates, they have few tax treaties. Bermuda and Cayman have no tax treaties within the 70 countries. Guernsey has only seven tax treaties. Thus, Bermuda, Cayman, and Guernsey appear to be pass-through countries in case (*S*), simply because they impose no CIT and no WHT on dividends. However, in case (*R*), common pass-through countries tend to have large numbers of tax treaties. For instance, as in Table 6, the United Kingdom (GB) has 60 tax treaties within the 70 countries. Tax treaties tend to influence the structures of tax-minimizing routes more profoundly in case (*R*) than in case (*S*).

Overall, tax-minimizing indirect routes tend to pass through countries with extensive treaty networks and no WHT in case (*R*), and pass through countries with no CIT and no WHT in case (*S*). Note that Luxembourg and the Netherlands are not listed as common pass-through countries in case (*R*) nor in case (*S*). This is probably because they can impose CIT and WHT at positive rates, as shown in Table 2. However, Luxembourg and the Netherlands may have favorable CIT systems for conduit entities, which are not considered in this paper. Moreover, Luxembourg and the Netherlands can have preferential tax treaties with certain countries. These features can make Luxembourg and the Netherlands attractive pass-through countries in real-world cases.

4 Empirical application

Multinational firms can choose investment (ownership) structures that influence foreign direct investment (FDI) between countries at an aggregate level. If such investors consider tax as an important factor when choosing investment structures, the results of the network analysis in Sect. 3 will be consistent with FDI data. For instance, Mintz

and Weichenrieder (2010), Lewellen and Robinson (2013) and Dyreng et al. (2015) discover that multinational firms use indirect ownership structures for tax motives.

Here, I investigate whether the network variables in Sect. 3 help explain the patterns of FDI by using Ordinary Least Squares (OLS) and Poisson Pseudo Maximum Likelihood (PPML) estimation methods. Silva and Tenreyro (2006) propose the PPML estimation to deal with numerous zero values in trade data. Because FDI data also exhibit a similar pattern, I adopt the PPML estimation to complement the log-linear OLS estimation.[22]

4.1 Data and variables

For a pair of countries i and j, FDI_{ij} is defined as the inward FDI stock from i to j in millions of US dollars. I use the bilateral FDI data of the UNCTAD that are compiled on the basis of immediate ownership relations.[23] Because the network variables are calculated without time-series dimension, I focus on cross-sectional data for year 2012 or the latest year available.[24] For instance, I use the FDI data of Luxembourg for year 2011 because the data for 2012 is unavailable. Because all the 70 countries except for Gibraltar (GI) and Guernsey (GG) are included in the bilateral FDI data, there are 4556 ($= 68 \times 67$) pairs of countries.

Table 10 provides descriptive statistics for regression variables. Note that the minimum value of FDI_{ij} is negative and there are 86 observations with negative FDI_{ij} in the data.[25] Also, there are 2141 observations with zero FDI_{ij}. To keep the observations with zero FDI_{ij}, the natural logarithm of inward FDI stock, denoted by $\ln \text{FDI}_{ij}$, is calculated as $\ln(\text{FDI}_{ij} + 1)$. Since the log-linear OLS and the PPML estimations exclude the observations with negative FDI_{ij}, there remain 4470 observations in regressions.

Three groups of network variables, DIRECT, TSHOP, and PASS, are introduced and calculated in Sect. 3. These network variables are defined for a pair of countries i and j in each case of (R) and (S).

Firstly, each DIRECT variable is an indicator variable for the existence of a tax-minimizing direct route. In case (R), if there is a tax-minimizing direct route from i to j, $\text{DIRECT}_{ij} = 1$, and otherwise, $\text{DIRECT}_{ij} = 0$. Also, DIRECTS_{ij} indicates the existence of a tax-minimizing direct route from i to j in case (S).

[22] FDI into a country may depend on the FDI in proximate countries. To control for such spatial interdependence, Blonigen et al. (2007) use an autoregressive model of FDI with spatial lags and find significant spatial effects.

[23] According to UNCTAD (2009), inward FDI flows (and stocks) should be classified by the residence country of immediate foreign direct investors. For instance, if an American parent firm invests in a Chinese subsidiary firm through an intermediate subsidiary firm in Hong Kong, this investment is classified as the FDI from the United States to Hong Kong and as the FDI from Hong Kong to China.

[24] It often takes considerable time (e.g., 10 years) to negotiate and pass a tax treaty. Once in force, a tax treaty often lasts for a long period of time (e.g., 30 years) without any amendment. There are limited changes in tax treaties during the period between 2012 and 2015. However, if there are any, changes in tax treaties during the period may weaken the analysis in this section. Later, for a robustness check, I use the FDI data for year 2015 from an alternative source and find consistent results.

[25] Negative FDI may indicate reverse investment or disinvestment.

Table 10 Descriptive statistics for regression variables

Variable	Obs	Mean	SD	Min	Max
FDI_{ij}	4556	4146.52	23,234.57	− 12,850.94	592,273.20
$DIRECT_{ij}$	4556	0.61	0.49	0	1
$DIRECTS_{ij}$	4556	0.75	0.43	0	1
$TSHOP_{ij}$	4556	3.58	6.09	0	34
$TSHOPS_{ij}$	4556	2.21	4.85	0	30
$PASS_{ij}$	4556	1.04	4.27	0	100
$PASSS_{ij}$	4556	2.09	11.06	0	100
$CONTIG_{ij}$	4556	0.03	0.18	0	1
$COMLANG_{ij}$	4556	0.09	0.29	0	1
$COMLEG_{ij}$	4556	0.25	0.43	0	1
$COLONY_{ij}$	4556	0.03	0.16	0	1
$DIST_{ij}$	4556	7.45	4.87	0.16	19.65
RTA_{ij}	4556	0.48	0.50	0	1
CIT_i	4556	21.67	7.86	0	35
GDP_i	4556	1,028,373.00	2,312,482.00	3410.10	16,155,254.80
FTC_i	4556	0.46	0.50	0	1
$SFTC_{ij}$	4556	0.17	0.38	0	1

I expect that each DIRECT variable is positively related to inward FDI stock. When the direct route is tax-minimizing between a pair of residence and source countries, investors will consider the direct route more attractive than indirect routes, and thus, they will use the direct route more frequently. Hong (2014) develops a game-theoretic model of treaty shopping and tax auditing between an investor and a tax agency, where the existence of a tax-minimizing direct route increases the investor's equilibrium probability to use the direct route. This equilibrium behavior leads to increased FDI between residence and source countries at an aggregate level.

Secondly, each TSHOP variable is a continuous variable specifying the difference between the foreign tax rates of the direct route and a tax-minimizing route. In case (R), the treaty shopping rate among routes from i to j is defined as $\text{TSHOP}_{ij} = t_{ji} - f(r^*)$, where r^* is a tax-minimizing route from i to j. Also, TSHOPS_{ij} denotes the treaty shopping rate among routes from i to j in case (S).

I expect that each TSHOP variable is negatively related to inward FDI stock. When the direct route is not tax-minimizing between a pair of residence and source countries, a TSHOP variable takes a positive value. Moreover, as the TSHOP variable increases, by definition, the difference becomes larger between the foreign tax rates of the direct route and a tax-minimizing route. Thus, investors will consider the direct route less attractive than tax-minimizing indirect routes. This can reduce FDI between residence and source countries at an aggregate level.

Thirdly, each PASS variable is a continuous variable specifying the relative frequency of being a pass-through country in tax-minimizing indirect routes. In case

(R), the pass-through rate of i to j, denoted by PASS_{ij}, is defined as the percentage of tax-minimizing indirect routes to j passing through i among all tax-minimizing indirect routes to j. Also, PASSS_{ij} denotes the pass-through rate of i to j in case (S).

I expect that each PASS variable is positively related to inward FDI stock. If there is a tax-minimizing indirect route between a pair of residence and source countries, investors will use the indirect route by funnelling their investments via a pass-through country. Thus, the use of an indirect route can increase FDI between a pass-through country and a source country. If there is more than one tax-minimizing indirect route, indirect investments can be distributed over possible pass-through countries. However, if a country is more likely to be a pass-through country in tax-minimizing indirect routes to a source country, the portion of indirect investments via the country can be larger than those via other countries.

There are six bilateral (pair-specific) variables to describe the relationship between countries i and j. CONTIG_{ij}, COMLANG_{ij}, COMLEG_{ij}, and COLONY_{ij} are indicator variables for a shared border, a common official language, a common legal origin, and a colonial relationship, respectively. DIST_{ij} is defined as the population-weighted distance between countries i and j in thousands of kilometers. Head et al. (2010) use these bilateral variables to examine the patterns of international trade. I obtain the data for these variables from the CEPII.[26] In addition, RTA_{ij} is an indicator variable for the existence of a regional trade agreement (RTA) between countries i and j. Egger and Larch (2008) examine the determinants of a regional trade agreement. I obtain the RTA data from Mario Larch's database.[27]

Corporate income tax rates and gross domestic product are variables specific to a country, depending on whether it is a residence country or a source country. CIT_i and CIT_j denote corporate income tax rates in residence country i and source country j, respectively. Each CIT rate is in percentage and from Deloitte International Tax Source and PwC Worldwide Tax Summaries. GDP_i and GDP_j denote gross domestic product at current prices in millions of US dollars in residence country i and source country j, respectively. Because I focus on a cross-sectional analysis, I use GDP data for year 2012 from the United Nations Statistics Division.[28] For my regression analysis I use the natural logarithm of GDP.[29]

In addition, I consider two variables about foreign tax credit (FTC) systems to control for their effects on FDI. These FTC variables are constructed with the information on tax laws from Deloitte International Tax Source and PwC Worldwide Tax Summaries.

First, FTC_i is a country-specific indicator variable. In country i, if there is a foreign tax credit system for dividends from foreign subsidiary firms, $\text{FTC}_i = 1$, and $\text{FTC}_i = 0$ otherwise. If country i has an exemption system for dividends from foreign subsidiaries, I assume that $\text{FTC}_i = 0$. Notably, all the capital-exporting residence

[26] Accessed at cepii.fr/CEPII/en/bdd_modele/bdd.asp.

[27] Accessed at ewf.uni-bayreuth.de/en/research/RTA-data.

[28] Accessed at unstats.un.org/unsd/snaama/dnlList.asp.

[29] For residence country i, $\ln \text{GDP}_i$ is calculated as $\ln(\text{GDP}_i + 1)$, and similarly, for source country j.

countries in Table 5 except for Ireland (IE), Korea (KR), and the United States (US) have such exemption systems.

Second, SFTC_{ij} is a pair-specific indicator variable. If there is a sufficient FTC in country i for taxes paid in country j, $\text{SFTC}_{ij} = 1$, and $\text{SFTC}_{ij} = 0$ otherwise. Precisely, if $\text{FTC}_i = 1$ and if $\text{CIT}_i > \text{CIT}_j + t_{ji}(1 - \text{CIT}_j/100)$, $\text{SFTC}_{ij} = 1$, and $\text{SFTC}_{ij} = 0$ otherwise. If $\text{SFTC}_{ij} = 1$, CIT and WHT paid in country j are fully creditable against tax liabilities in country i.[30]

I also use home and source dummy variables to control for unobserved country-specific factors. When doing so, I replace country-specific variables, CIT, GDP, and FTC, with home and source dummy variables.

4.2 Results

Now, I present regression results with network variables. Table 11 provides the results of the OLS regressions with network variables DIRECT_{ij}, TSHOP_{ij}, and PASS_{ij} in case (R). In the OLS regressions, the dependent variable is $\ln \text{FDI}_{ij}$, the natural logarithm of inward FDI stock. Each regression includes one of the three network variables as a main independent variable.[31] Each regression also includes bilateral variables CONTIG_{ij}, COMLANG_{ij}, COMLEG_{ij}, COLONY_{ij}, DIST_{ij}, and RTA_{ij} to control for the relationship between countries i and j. In addition, to consider the effects of foreign tax credit systems, columns (1)–(3) include a home-specific indicator variable FTC_i. Columns (4)–(6) include a pair-specific indicator variable SFTC_{ij}. Furthermore, to control for country-specific factors, columns (1)–(3) include CIT and log-scaled GDP in home country i and source country j. Columns (4)–(6) include dummy variables for home and source countries.

Table 11 shows that each of the three network variables DIRECT_{ij}, TSHOP_{ij}, and PASS_{ij} is significantly related to inward FDI stock.

In columns (1) and (4), the coefficients on DIRECT_{ij} are positive and significant at the 1% level. Column (4) suggests that the inward FDI stock via a tax-minimizing direct route is 2.14 ($= \exp(0.76)$) times larger than the inward FDI stock via a direct route that is not tax-minimizing.

In columns (2) and (5), the coefficients on TSHOP_{ij} are negative and significant at the 1% level. Column (5) suggests that the inward FDI stock decreases by 3.92% ($= \exp(-0.04) - 1$) as the treaty shopping rate increases by one percentage point.

In columns (3) and (6), the coefficients on PASS_{ij} are positive and significant at the 1% level. Column (6) suggests that the inward FDI stock increases by 3.05% ($= \exp(0.03) - 1$) as the pass-through rate increases by one percentage point.

[30] If $\text{SFTC}_{ij} = 1$, the direct route $i \rightarrow j$ yields the same net-of-tax income as a tax-minimizing indirect route, and the investor may have no incentive to use an indirect route. Thus, one might expect a positive relationship between FDI_{ij} and SFTC_{ij}. However, I could not find a clear relationship between the two variables. In Tables 11, 12, 13 and 14, the coefficients on SFTC_{ij} show mixed signs and become insignificant in most specifications.

[31] I find that DIRECT_{ij} and TSHOP_{ij} are strongly and negatively correlated with each other. The Pearson correlation coefficient between the two variables is about -0.73, while the Spearman coefficient is -0.96. To avoid issues related to multicollinearity, I do not use DIRECT_{ij} and TSHOP_{ij} together in the regressions.

Table 12 provides the results of the OLS regressions with network variables $DIRECTS_{ij}$, $TSHOPS_{ij}$, and $PASSS_{ij}$ in case (*S*). In columns (1) and (4), the coefficients on $DIRECTS_{ij}$ are positive and significant at the 1% level. In columns (2) and (5), the coefficients on $TSHOPS_{ij}$ are negative and significant at the 1% level. However, in columns (3) and (6), the coefficients on $PASSS_{ij}$ are insignificant.

Tables 13 and 14 provide the results of the PPML regressions with network variables in cases (*R*) and (*S*), respectively. In the PPML regressions, FDI_{ij} is the dependent variable and the observations with negative FDI_{ij} are dropped. Tables 13 and 14 show that $DIRECT_{ij}$ and $DIRECTS_{ij}$ are positively and significantly related to inward FDI stock. Moreover, $TSHOP_{ij}$ and $TSHOPS_{ij}$ are negatively and significantly related to inward FDI stock. By using home and source dummies to control for country-specific factors, in column (6), I find that $PASS_{ij}$ and $PASSS_{ij}$ are positively and significantly related to inward FDI stock.

The Coordinated Direct Investment Survey (CDIS) of the International Monetary Fund (IMF) provides an alternative source of bilateral FDI data. To check the robustness of my analysis, I use the FDI data for year 2015 or the latest year available from the CDIS and present the results of the OLS regressions in Appendix A. As before, the dependent variable is the natural logarithm of inward FDI stock. Each regression includes one of the network variables in cases (*R*) and (*S*) as a main independent variable. Each regression also includes bilateral control variables, together with home and source dummies.

Table 16 shows that all the network variables except for $PASSS_{ij}$ are significantly related to inward FDI stock. The estimated coefficients are positive for $DIRECT_{ij}$ and $DIRECTS_{ij}$, negative for $TSHOP_{ij}$ and $TSHOPS_{ij}$, and positive for $PASS_{ij}$ and $PASSS_{ij}$. These results are consistent with those in Tables 11 and 12 based on the FDI data of the UNCTAD.

Overall, I find that the existence of a tax-minimizing direct route is positively and significantly related to inward FDI stock.[32] This observation is consistent with a game-theoretic model of treaty shopping (Hong 2014). Therefore, by making a direct route tax-minimizing, countries can encourage foreign investment via the direct route.

I also find that the treaty shopping rate is negatively and significantly related to inward FDI stock. Therefore, even if a direct route is not tax-minimizing, by reducing the treaty shopping rate, countries can encourage foreign investment via the direct route. For instance, to reduce the treaty shopping rate, countries can unilaterally reduce withholding tax rates on dividends paid to foreign investors.

In addition, I find some evidence that the pass-through rate is positively related to inward FDI stock. This relationship tends to be statistically significant in case (*R*), where pass-through countries are assumed to impose CIT at reduced rates. Increased FDI from pass-through countries may result partly from treaty shopping, i.e., the use

[32] I admit the possibility that an unobserved factor can affect both FDI and the structure of tax-minimizing routes. For instance, if there is a cost of treaty negotiation for low tax rates, countries may incur the cost only when they expect large FDI. This can also explain the positive relationship between FDI and the existence of a tax-minimizing direct route. However, in this paper, I used the bilateral variables of Head et al. (2010) that describe the relationship between countries to control for unobserved factors, such as treaty negotiation process and cost. In this context, I believe that a potential endogeneity problem (due to unobserved factors) is minimized.

Table 11 FDI and network variables in case (*R*)

	(1) $\ln FDI_{ij}$	(2) $\ln FDI_{ij}$	(3) $\ln FDI_{ij}$	(4) $\ln FDI_{ij}$	(5) $\ln FDI_{ij}$	(6) $\ln FDI_{ij}$
$DIRECT_{ij}$	0.89***			0.76***		
	(0.09)			(0.12)		
$TSHOP_{ij}$		− 0.05***			− 0.04***	
		(0.01)			(0.01)	
$PASS_{ij}$			0.05***			0.03***
			(0.01)			(0.01)
$CONTIG_{ij}$	1.26***	1.28***	1.36***	1.37***	1.40***	1.43***
	(0.26)	(0.26)	(0.26)	(0.21)	(0.21)	(0.21)
$COMLANG_{ij}$	2.20***	2.26***	2.30***	1.19***	1.20***	1.21***
	(0.20)	(0.20)	(0.20)	(0.17)	(0.17)	(0.18)
$COMLEG_{ij}$	0.35***	0.34***	0.34***	0.56***	0.57***	0.57***
	(0.11)	(0.11)	(0.11)	(0.09)	(0.09)	(0.09)
$COLONY_{ij}$	0.46	0.45	0.44	0.57**	0.54*	0.55**
	(0.34)	(0.34)	(0.34)	(0.28)	(0.28)	(0.28)
$DIST_{ij}$	− 0.15***	− 0.15***	− 0.16***	− 0.20***	− 0.20***	− 0.21***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
RTA_{ij}	0.59***	0.65***	0.69***	0.28***	0.33***	0.37***
	(0.10)	(0.09)	(0.09)	(0.10)	(0.10)	(0.10)
$SFTC_{ij}$				− 0.02	− 0.02	0.07
				(0.13)	(0.13)	(0.13)
CIT_i	− 0.04***	− 0.04***	− 0.05***			
	(0.01)	(0.01)	(0.01)			
FTC_i	− 1.15***	− 1.17***	− 1.16***			
	(0.09)	(0.09)	(0.09)			
$\ln GDP_i$	0.92***	0.90***	0.93***			
	(0.03)	(0.03)	(0.03)			
CIT_j	0.02**	0.01**	0.01**			
	(0.01)	(0.01)	(0.01)			
$\ln GDP_j$	0.59***	0.58***	0.55***			
	(0.03)	(0.03)	(0.03)			
Home dummy	No	No	No	Yes	Yes	Yes
Source dummy	No	No	No	Yes	Yes	Yes
Constant	− 14.70***	− 13.61***	− 13.69***	2.99***	3.49***	3.10***
	(0.50)	(0.50)	(0.50)	(0.36)	(0.37)	(0.36)
Observations	4470	4470	4470	4470	4470	4470
R-squared	0.38	0.37	0.37	0.66	0.66	0.66

Robust standard errors in parentheses; *significant at 10%; ** at 5%; *** at 1%

Table 12 FDI and network variables in case (*S*)

	(1) $\ln FDI_{ij}$	(2) $\ln FDI_{ij}$	(3) $\ln FDI_{ij}$	(4) $\ln FDI_{ij}$	(5) $\ln FDI_{ij}$	(6) $\ln FDI_{ij}$
$DIRECTS_{ij}$	0.78***			0.48***		
	(0.10)			(0.11)		
$TSHOPS_{ij}$		− 0.05***			− 0.04***	
		(0.01)			(0.01)	
$PASSS_{ij}$			0.01			0.01
			(0.01)			(0.01)
$CONTIG_{ij}$	1.28***	1.30***	1.33***	1.41***	1.40***	1.42***
	(0.26)	(0.26)	(0.26)	(0.21)	(0.21)	(0.21)
$COMLANG_{ij}$	2.25***	2.27***	2.30***	1.21***	1.19***	1.21***
	(0.20)	(0.20)	(0.20)	(0.17)	(0.17)	(0.18)
$COMLEG_{ij}$	0.35***	0.35***	0.33***	0.57***	0.57***	0.57***
	(0.11)	(0.11)	(0.11)	(0.09)	(0.09)	(0.09)
$COLONY_{ij}$	0.48	0.45	0.47	0.56**	0.54*	0.55**
	(0.34)	(0.34)	(0.34)	(0.28)	(0.28)	(0.28)
$DIST_{ij}$	− 0.15***	− 0.15***	− 0.16***	− 0.20***	− 0.20***	− 0.21***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
RTA_{ij}	0.67***	0.70***	0.71***	0.33***	0.35***	0.38***
	(0.09)	(0.09)	(0.09)	(0.10)	(0.10)	(0.10)
$SFTC_{ij}$				0.002	− 0.01	0.10
				(0.13)	(0.13)	(0.13)
CIT_i	− 0.04***	− 0.04***	− 0.04***			
	(0.01)	(0.01)	(0.01)			
FTC_i	− 1.16***	− 1.18***	− 1.21***			
	(0.09)	(0.09)	(0.09)			
$\ln GDP_i$	0.91***	0.91***	0.93***			
	(0.03)	(0.03)	(0.03)			
CIT_j	0.01**	0.01**	0.01**			
	(0.01)	(0.01)	(0.01)			
$\ln GDP_j$	0.58***	0.57***	0.55***			
	(0.03)	(0.03)	(0.03)			
Home dummy	No	No	No	Yes	Yes	Yes
Source dummy	No	No	No	Yes	Yes	Yes
Constant	− 14.46***	− 13.62***	− 13.77***	2.95***	3.38***	3.29***
	(0.50)	(0.50)	(0.50)	(0.36)	(0.37)	(0.56)
Observations	4470	4470	4470	4470	4470	4470
R-squared	0.37	0.37	0.36	0.66	0.66	0.66

Robust standard errors in parentheses; *significant at 10%; ** at 5%; *** at 1%

Table 13 PPML estimation for FDI and network variables in case (*R*)

	(1) FDI_{ij}	(2) FDI_{ij}	(3) FDI_{ij}	(4) FDI_{ij}	(5) FDI_{ij}	(6) FDI_{ij}
$DIRECT_{ij}$	1.18***			0.59***		
	(0.14)			(0.18)		
$TSHOP_{ij}$		− 0.15***			− 0.08***	
		(0.02)			(0.01)	
$PASS_{ij}$			0.01			0.01**
			(0.01)			(0.01)
$CONTIG_{ij}$	0.37*	0.38*	0.46**	0.62***	0.58***	0.65***
	(0.21)	(0.22)	(0.22)	(0.13)	(0.13)	(0.14)
$COMLANG_{ij}$	1.36***	1.37***	1.45***	0.54***	0.59***	0.56***
	(0.18)	(0.19)	(0.18)	(0.14)	(0.13)	(0.13)
$COMLEG_{ij}$	0.06	0.07	0.05	0.12	0.13	0.09
	(0.19)	(0.20)	(0.19)	(0.10)	(0.10)	(0.10)
$COLONY_{ij}$	0.12	0.14	0.21	0.40***	0.33***	0.43***
	(0.19)	(0.19)	(0.20)	(0.11)	(0.12)	(0.12)
$DIST_{ij}$	− 0.13***	− 0.12***	− 0.13***	− 0.14***	− 0.14***	− 0.15***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
RTA_{ij}	− 0.05	− 0.01	0.12	0.28**	0.29**	0.33***
	(0.16)	(0.16)	(0.15)	(0.12)	(0.12)	(0.13)
$SFTC_{ij}$				0.24	0.20	0.36*
				(0.21)	(0.21)	(0.20)
CIT_i	− 0.01	− 0.004	− 0.001			
	(0.01)	(0.01)	(0.01)			
FTC_i	− 0.75***	− 0.78***	− 0.87***			
	(0.11)	(0.11)	(0.12)			
$\ln GDP_i$	0.65***	0.64***	0.66***			
	(0.06)	(0.06)	(0.06)			
CIT_j	0.01	0.01	0.005			
	(0.01)	(0.01)	(0.01)			
$\ln GDP_j$	0.61***	0.60***	0.58***			
	(0.06)	(0.06)	(0.06)			
Home dummy	No	No	No	Yes	Yes	Yes
Source dummy	No	No	No	Yes	Yes	Yes
Constant	− 8.48***	− 7.29***	− 7.65***	4.55***	9.24***	9.06***
	(1.03)	(1.02)	(1.08)	(0.54)	(0.23)	(0.24)
Observations	4470	4470	4470	4470	4470	4470
R-squared	0.42	0.43	0.39	0.77	0.78	0.76

Robust standard errors in parentheses; *significant at 10%; ** at 5%; *** at 1%

Table 14 PPML estimation for FDI and network variables in case (*S*)

	(1) FDI_{ij}	(2) FDI_{ij}	(3) FDI_{ij}	(4) FDI_{ij}	(5) FDI_{ij}	(6) FDI_{ij}
$DIRECTS_{ij}$	1.32***			0.78***		
	(0.15)			(0.17)		
$TSHOPS_{ij}$		− 0.17***			− 0.11***	
		(0.02)			(0.02)	
$PASSS_{ij}$			0.003			0.04***
			(0.004)			(0.01)
$CONTIG_{ij}$	0.37*	0.40*	0.45**	0.57***	0.56***	0.62***
	(0.22)	(0.22)	(0.22)	(0.13)	(0.13)	(0.14)
$COMLANG_{ij}$	1.43***	1.40***	1.46***	0.61***	0.62***	0.59***
	(0.19)	(0.19)	(0.18)	(0.13)	(0.13)	(0.13)
$COMLEG_{ij}$	0.03	0.06	0.05	0.10	0.12	0.09
	(0.20)	(0.20)	(0.19)	(0.10)	(0.10)	(0.10)
$COLONY_{ij}$	0.19	0.20	0.23	0.39***	0.34***	0.41***
	(0.19)	(0.19)	(0.20)	(0.12)	(0.12)	(0.12)
$DIST_{ij}$	− 0.12***	− 0.12***	− 0.13***	− 0.14***	− 0.14***	− 0.15***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
RTA_{ij}	0.05	0.08	0.13	0.28**	0.29**	0.31**
	(0.16)	(0.15)	(0.15)	(0.12)	(0.12)	(0.13)
$SFTC_{ij}$				0.26	0.27	0.37*
				(0.22)	(0.22)	(0.20)
CIT_i	− 0.0002	− 0.0008	− 0.00002			
	(0.01)	(0.01)	(0.01)			
FTC_i	− 0.82***	− 0.83***	− 0.89***			
	(0.11)	(0.11)	(0.12)			
$\ln GDP_i$	0.65***	0.64***	0.66***			
	(0.06)	(0.06)	(0.06)			
CIT_j	0.005	0.005	0.01			
	(0.01)	(0.01)	(0.01)			
$\ln GDP_j$	0.60***	0.59***	0.58***			
	(0.06)	(0.06)	(0.06)			
Home dummy	No	No	No	Yes	Yes	Yes
Source dummy	No	No	No	Yes	Yes	Yes
Constant	− 8.77***	− 7.32***	− 7.73***	3.99***	6.60***	4.77***
	(1.05)	(1.05)	(1.09)	(0.58)	(0.36)	(0.53)
Observations	4470	4470	4470	4470	4470	4470
R-squared	0.41	0.40	0.38	0.78	0.78	0.76

Robust standard errors in parentheses; *significant at 10%; ** at 5%; *** at 1%

of indirect routes via the pass-through countries. Therefore, source countries can use the pass-through rate as an index to monitor treaty shopping.

5 Conclusion

In this paper, I construct a tax rate matrix to represent a tax treaty network and develop a computation algorithm to examine the structure of tax-minimizing routes. I find that the use of a tax-minimizing indirect route can substantially reduce tax on dividends incurred by a multinational investor. I also find that the existence of a tax-minimizing direct route is positively related to foreign investment via the direct route.

When a direct route is not tax-minimizing, to prevent the use of a tax-minimizing indirect route, i.e., to prevent treaty shopping, countries need to consider negotiating a new tax treaty to make the direct route tax-minimizing. These countries may not necessarily experience tax revenue loss, as the new treaty makes the direct route tax-minimizing and attracts more foreign investment.

For future studies, it will be interesting to study the relationship between tax treaties and ownership structures of multinational firms. In this paper, I focused on FDI data to see the plausibility of the treaty network analysis. However, the aggregate FDI data do not reveal the actual ownership structures (i.e., investment routes) of multinational firms. Hence, to check the plausibility of the network analysis, it is important to examine whether and how multinational firms organize indirect ownership structures when direct ownership structures are not tax-minimizing.

It will also be interesting to study the network effect of LOB (limitation on benefits) provisions in tax treaties on the structure of tax-minimizing investment routes. In this paper, when constructing the tax rate matrix, I considered a restrictive version of LOB provisions that only allow certain pension funds to obtain the benefit of the lowest withholding tax rate. However, in real-world tax treaties, LOB provisions can impose various residency requirements. Thus, it is important to examine how LOB provisions with various residency requirements affect the structure of tax-minimizing investment routes in tax treaty networks.

Lastly, it will be interesting to study indirect structures that reduce tax on interest and royalties. In this paper, I concentrated on withholding tax rates on dividends to describe a tax treaty network. However, tax treaties also affect withholding tax rates on interest and royalties. Moreover, FDI can take the form of debt as well as equity. FDI with debt instruments may be influenced by tax rates on interest as FDI with equity instruments may be influenced by tax rates on dividends. Therefore, it is important to examine how tax treaties affect indirect structures to reduce tax on interest and royalties and how such indirect structures affect FDI patterns.

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Appendix A: Tables

Table 15 provides examples of tax-minimizing routes between selected pairs of home and source countries. According to the FDI data of the UNCTAD, the United States (US), Japan (JP), France (FR), Germany (DE), and the United Kingdom (GB) are the countries with the largest net outward FDI stock, and selected as home countries. China (CN), Mexico (MX), Indonesia (ID), Poland (PL), and Saudi Arabia (SA) are the countries with the largest net inward FDI stock, and selected as source countries. It is worthwhile to note that every tax-minimizing route passes through at most one country between the pairs of home and source countries in Table 15.

Table 16 provides the results of the OLS regressions with the FDI data from the CDIS. Overall, these results are consistent with those in Tables 11 and 12 based on the FDI data of the UNCTAD.

Table 15 Examples of tax-minimizing routes in case (*R*)

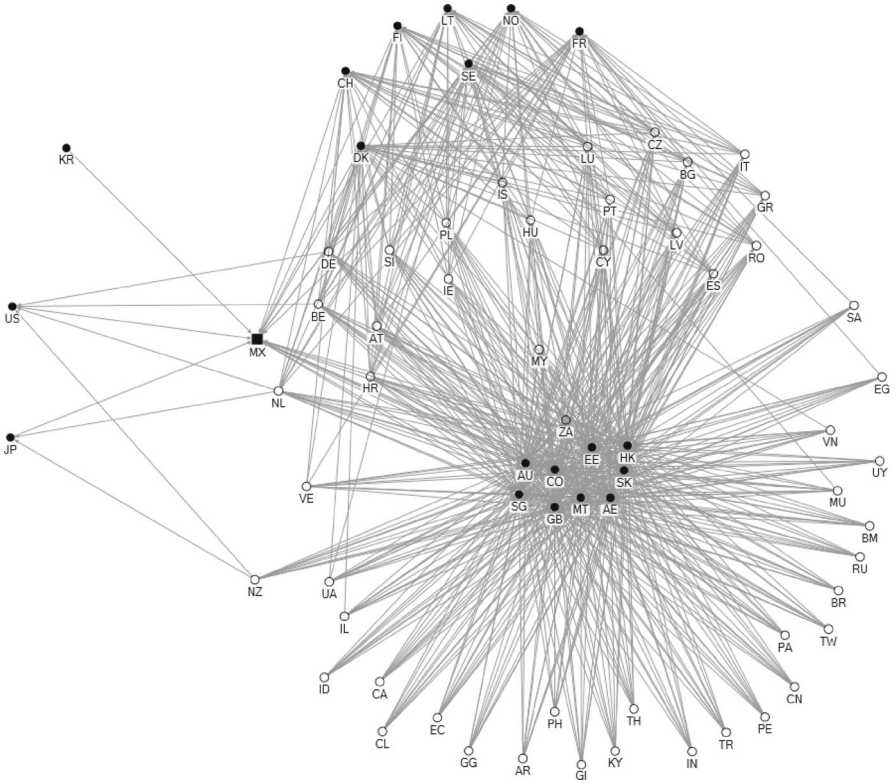
Home		Pass-through		Source	TSHOP
US	→	BE, DK, EC, EE, FI, FR, DE, HK, IE, LV, MT, MU, MX, NL, SG, SE, GB, VE	→	CN	4.05
US		→		MX	0
US	→	HK	→	ID	4.05
US	→	BE, CY, DK, EE, FI, FR, DE, HU, IE, IT,LV, MT, NL, SK, SE, AE, GB	→	PL	4.00
US	→	FR, IE	→	SA	4.00
JP	→	BE, EC, EE, FR, HK, IE, LV, MT, MU, MX,NL, SG, SE, CH, GB, VE	→	CN	4.05
JP		→		MX	0
JP	→	HK	→	ID	4.05
JP	→	BE, CY, EE, FR, HU, IE, LV, MT, NL, SK,SE, CH, AE, GB	→	PL	9.00
JP	→	FR, IE	→	SA	4.00
FR		→		CN	0
FR		→		MX	0
FR	→	HK	→	ID	4.05
FR		→		PL	0
FR		→		SA	0
DE		→		CN	0
DE	→	AU, CO, DK, EE, FI, FR, HK, LT, MT, NO,SG, SK, SE, CH, AE, GB, US	→	MX	4.00
DE	→	HK	→	ID	4.05
DE		→		PL	0
DE	→	FR, IE, ES	→	SA	4.00
GB		→		CN	0
GB		→		MX	0
GB	→	HK	→	ID	4.05
GB		→		PL	0
GB	→	FR, IE, ES	→	SA	4.00

Table 16 OLS estimation with FDI data from CDIS

	(1) $\ln FDI_{ij}$	(2) $\ln FDI_{ij}$	(3) $\ln FDI_{ij}$	(4) $\ln FDI_{ij}$	(5) $\ln FDI_{ij}$	(6) $\ln FDI_{ij}$
$DIRECT_{ij}$	1.01***					
	(0.13)					
$TSHOP_{ij}$		− 0.07***				
		(0.01)				
$PASS_{ij}$			0.02*			
			(0.01)			
$DIRECTS_{ij}$				0.72***		
				(0.11)		
$TSHOPS_{ij}$					− 0.07***	
					(0.01)	
$PASSS_{ij}$						0.01
						(0.01)
$CONTIG_{ij}$	1.35***	1.39***	1.43***	1.41***	1.39***	1.42***
	(0.19)	(0.20)	(0.20)	(0.20)	(0.20)	(0.20)
$COMLANG_{ij}$	0.89***	0.90***	0.92***	0.91***	0.89***	0.91***
	(0.16)	(0.16)	(0.17)	(0.16)	(0.16)	(0.17)
$COMLEG_{ij}$	0.37***	0.38***	0.39***	0.38***	0.39***	0.39***
	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)	(0.09)
$COLONY_{ij}$	1.13***	1.09***	1.10***	1.13***	1.10***	1.11***
	(0.26)	(0.26)	(0.27)	(0.26)	(0.26)	(0.27)
$DIST_{ij}$	− 0.16***	− 0.16***	− 0.17***	− 0.16***	− 0.16***	− 0.17***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
RTA_{ij}	0.35***	0.39***	0.47***	0.41***	0.43***	0.48***
	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)
$SFTC_{ij}$	0.00003	− 0.04	0.14	0.01	− 0.01	0.16
	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)
Home dummy	Yes	Yes	Yes	Yes	Yes	Yes
Source dummy	Yes	Yes	Yes	Yes	Yes	Yes
Constant	5.58***	6.61***	6.17***	5.75***	6.56***	5.83***
	(0.53)	(0.52)	(0.54)	(0.53)	(0.51)	(0.66)
Observations	4227	4227	4227	4227	4227	4227
R-squared	0.67	0.67	0.67	0.67	0.67	0.67

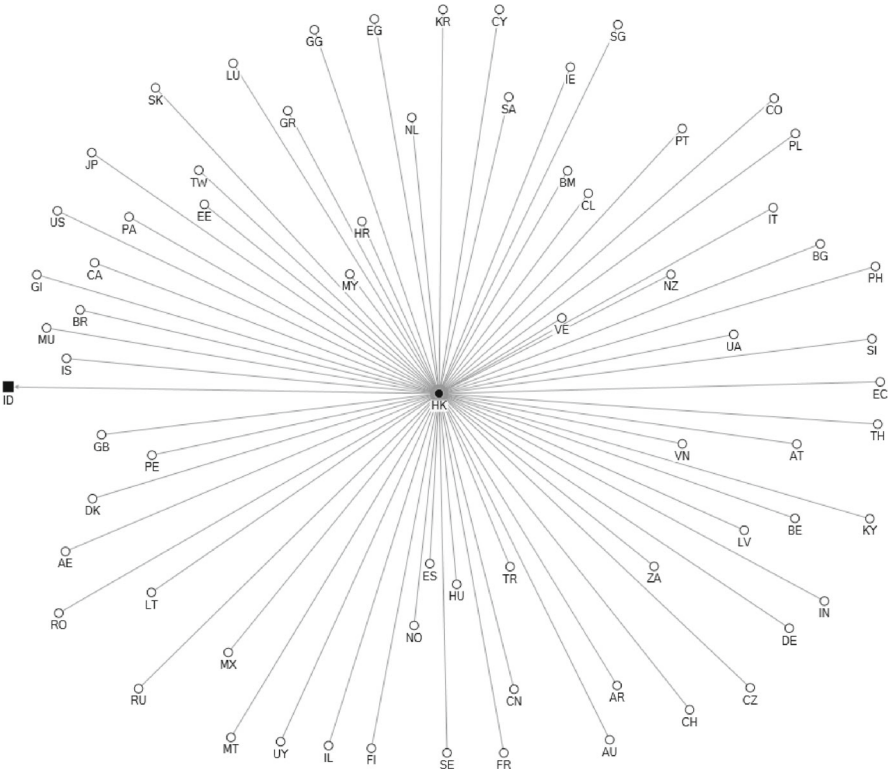
Robust standard errors in parentheses; *significant at 10%; ** at 5%; *** at 1%

Appendix B: Network diagrams



Note: ■ MX source; ● countries with tax-minimizing direct routes; ○ countries with no such routes

Fig. 3 Tax-minimizing routes to Mexico (MX)



Note: ■ ID source; ● countries with tax-minimizing direct routes;
○ countries with no such routes

Fig. 4 Tax-minimizing routes to Indonesia (ID)

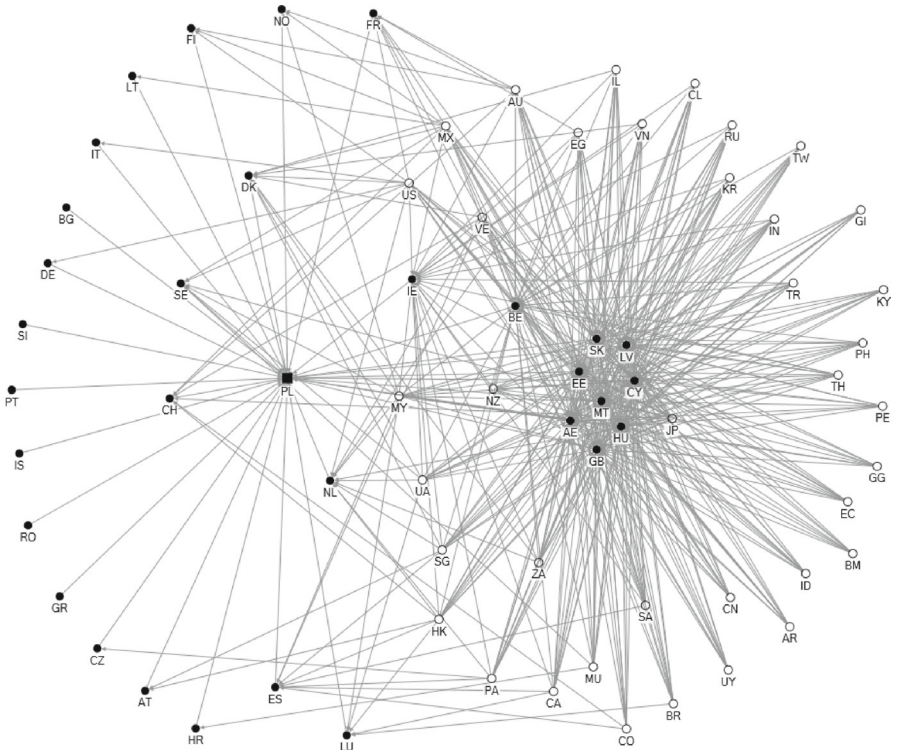
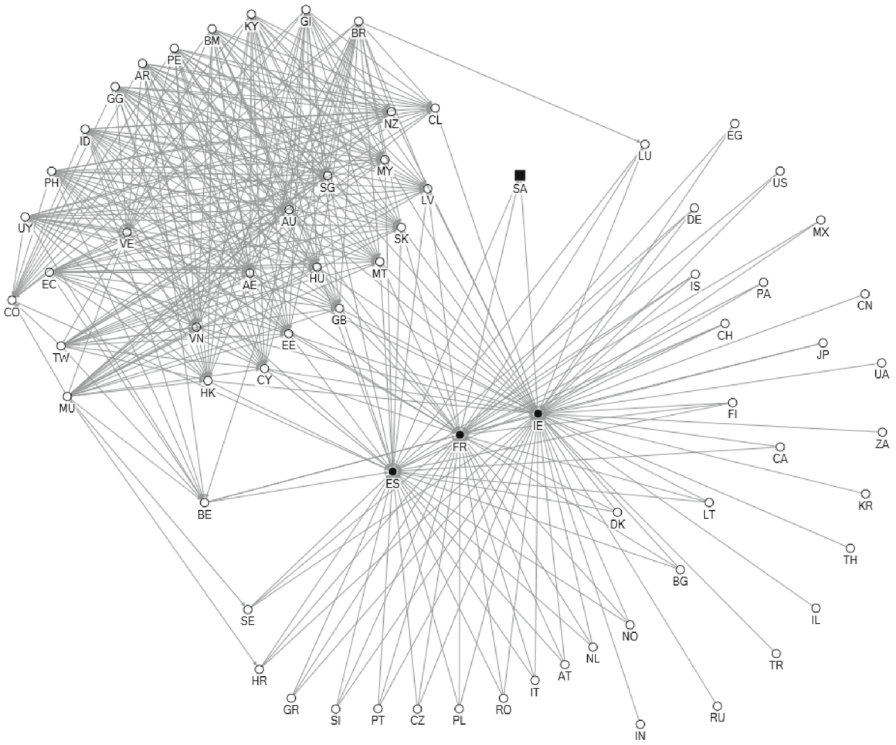
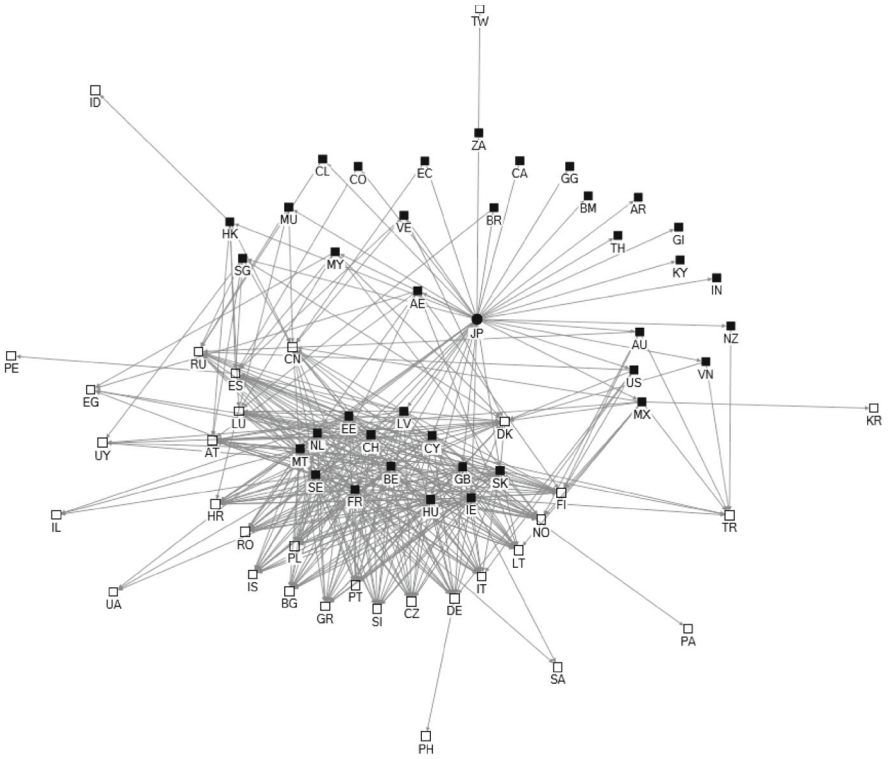


Fig. 5 Tax-minimizing routes to Poland (PL)



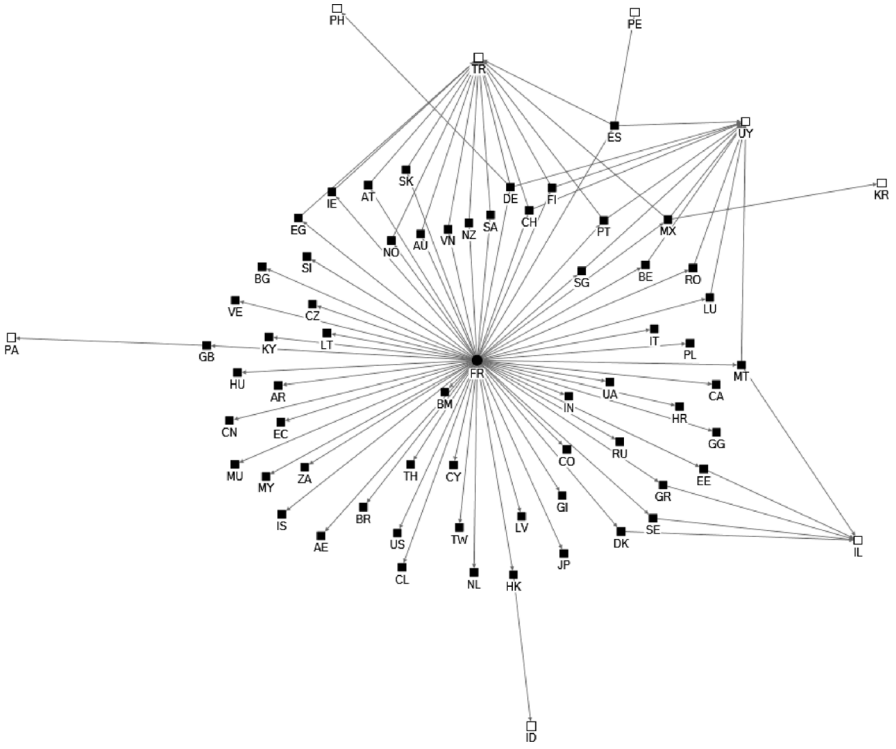
Note: ■ SA source; ● countries with tax-minimizing direct routes;
○ countries with no such routes

Fig. 6 Tax-minimizing routes to Saudi Arabia (SA)



Note: ● JP residence; ■ countries with tax-minimizing direct routes; □ countries with no such routes

Fig. 7 Tax-minimizing routes from Japan (JP)



Note: ● FR residence; ■ countries with tax-minimizing direct routes; □ countries with no such routes

Fig. 8 Tax-minimizing routes from France (FR)

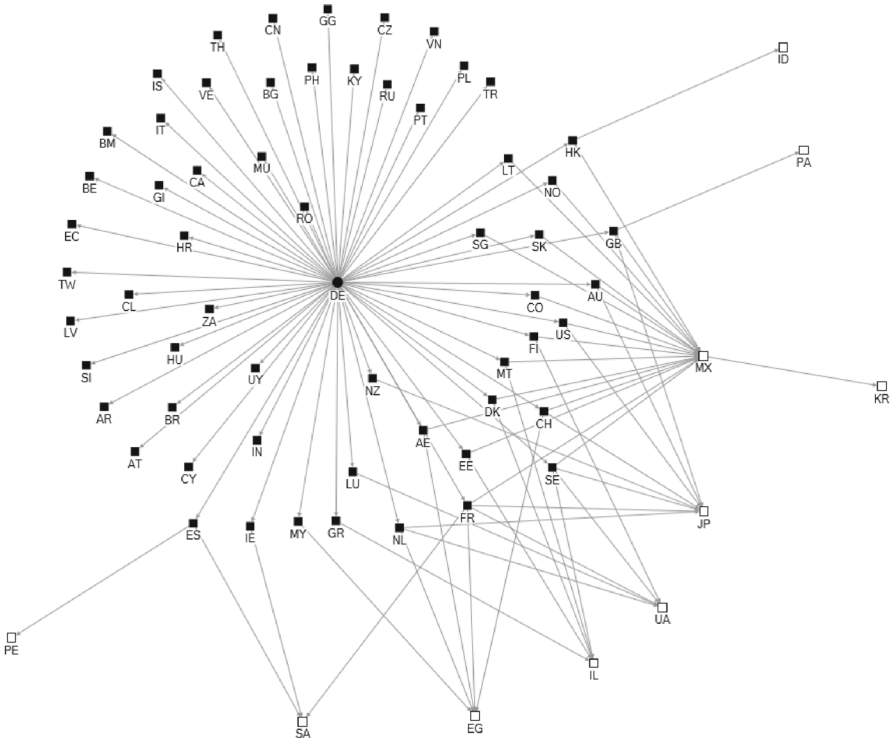
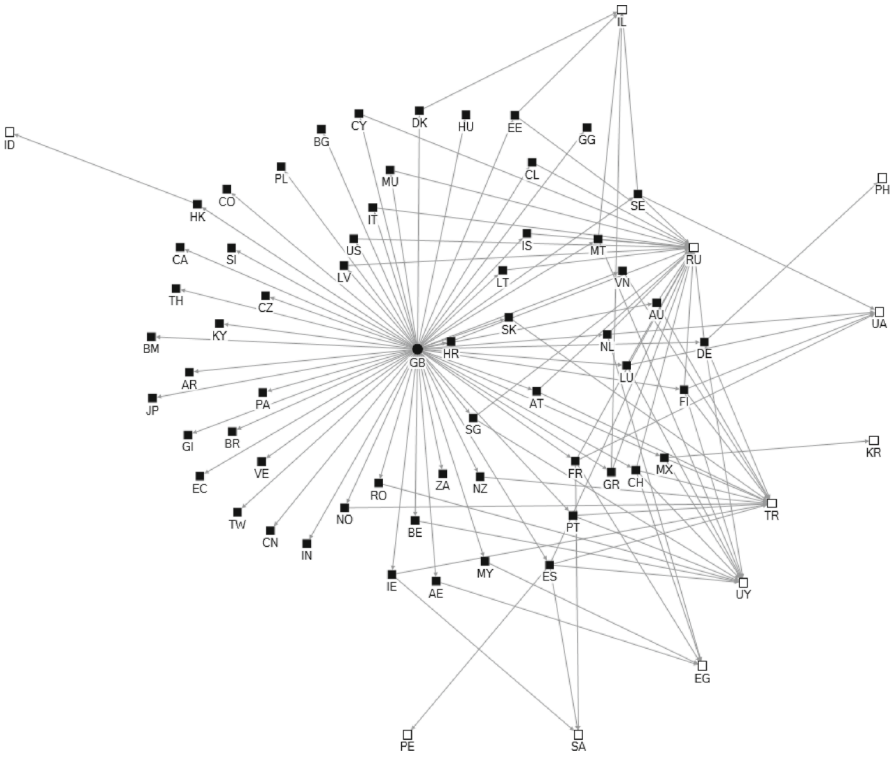


Fig. 9 Tax-minimizing routes from Germany (DE)



Note: ● GB residence; ■ countries with tax-minimizing direct routes; □ countries with no such routes

Fig. 10 Tax-minimizing routes from the United Kingdom (GB)

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